

1 Phonemic discovery in a polysynthetic
2 language: distributional and corpus evidence
3 for two additional vowels in Blackfoot

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1 Introduction

When documenting a language, a crucial early task is the phonemic analysis of the language to discover the contrastive sounds of a language.¹ A key test for contrast is lexical distinction via minimal pairs. However, this test can be difficult to apply in polysynthetic languages, which have phonologically and morphologically complex words and therefore fewer minimal pairs. This paper instead relies on evidence from phonological distribution and morphophonological alternations, as well as evidence from systematic gaps in a small corpus study. The techniques described here could be applied to other morphologically complex languages without many minimal pairs.

The empirical focus of the paper is Blackfoot (Algonquian; ISO 639-3: bla), which is described in a recent grammar (Frantz 2017) and dictionary (Frantz and Russell 2017; Genee and Junker 2018). I argue that Blackfoot has five contrastive long vowels /i: ε: a: ɔ: o:/ instead of the three /i: a: o:/ in traditional analyses. The evidence that the long mid vowels /ε:/ and /ɔ:/ are phonemic comes from their overlapping distribution with other long vowels as well as patterns in root alternations. Adding two new vowel phonemes to the inventory also has consequences for other aspects of the phonological system, such as the underlying forms of some roots. The reanalysis of the vowel inventory and lexicon is supported by a small corpus study of the entry headers in the most recent dictionary (Frantz and Russell 2017). Most of the data in this paper is from examples in Frantz and Russell (2017), with a phonetic transcription added (described in Section 3.4). Some words are from original fieldwork with Tootsinám (Beatrice Bullshields, abbreviated as “BB”).

The paper is organized as follows. In Section 2 I describe common tests for contrastivity and discuss reasons why each of these can be difficult to apply to polysynthetic languages. In Section 3 I discuss some aspects of Blackfoot phonology and

¹ Acknowledgements... The Blackfoot language belongs to the Blackfoot people. Any data cited from Tootsinám (Beatrice Bullshields) was collected 2011–2015 in Vancouver, BC and partially funded by the Jacobs Research Funds. Any mistakes in this paper are my own.

morphology which are necessary to understanding the rest of the paper. In Section 4 present arguments that long mid vowels are contrastive. In Section 5 I argue that roots which have been analyzed with an initial /w/ actually begin in long vowels, and that the [w] is epenthetic. Section 6 includes supporting arguments from a corpus study of the most recent dictionary. Section 7 returns to the question of how two phonemes remained undiscovered in Blackfoot. In Section 8 I conclude, and finish with implications of this paper for the synchronic and diachronic grammar of Blackfoot, as well as a description of a lexical database which can support future research into Blackfoot morphophonology.

2 Phonemic discovery in understudied polysynthetic languages

Phonemic analysis, once an important goal in its own right, is still an important step in the phonological analysis of any language (e.g., Bloomfield 1933, 1935; Sapir 1925; Swadesh 1934; Twaddell 1957 [1935]; Trubetzkoy 1969 [1939]; Pike 1947). Linguists use a number of different criteria to determine the phonemic units of a language (Steriade 2007; Drescher 2016; K.C.Hall 2013), and there are many different ideas about proper procedures for conducting phonemic analyses (e.g., Chao 1957 [1934]). However, some of these criteria and procedures are made even more difficult in understudied or polysynthetic languages. In this section I point out some of these factors.

2.1 Minimal pairs

The test of *lexical distinction* relies on minimal pairs. If substituting sound A for sound B changes the meaning of a word, then the sounds A and B contrast. This test establishes contrast by comparing allophones externally to other contrastive categories in the phonological system. Polysynthetic languages often have few minimal pairs, in part due to the length of the words: it is probabilistically unlikely that substituting a single sound will result in a second well-formed word. In contrast, for a language with monosyllabic words, it is relatively easy to create minimal pairs via substitution. This was pointed out succinctly by Swadesh (1936:299), and can even be quantified. The phonological neighborhood density of a word, w , which is the number of words that differ from word w by insertion, deletion, or substitution of a phoneme (Luce 1986; Vitevitch and Luce

1998). This number can approximate the number of minimal pairs, which are typically construed as the subset of neighborhood density that only includes substitution. As expected, neighborhood density is negatively correlated with word length: short words tend to have more neighbors than long words (Pisoni et al. 1985; Bard and Shillcock, 1993; Storkel 2004).

2.2 Allophones

There is a complement to the test of lexical distinction: if substitution does *not* change the meaning of the word, then A and B are non-contrastive variants of each other (e.g., free variants, dialectal variants, or contextual allophones). This test establishes *non*-contrast by comparing allophones internally within a family of non-contrastive sounds. Especially if there is a high degree of variability among allophones, a researcher might focus on establishing whether those allophones are non-contrastive. By nature though, this means that the researcher focuses on sounds that do not contrast, without necessarily establishing contrast with some other phonemic category.

2.3 Relationship to morphology

The relationship to morphology can be much more difficult to determine in polysynthetic languages than in morphologically simpler languages. There are many known cases of pseudo-minimal pairs, where a surface contrast is predictable once morphological boundaries are taken into consideration (e.g., Ladd 2006; Scobbie and Stuart-Smith 2008). But what if a full morphosyntactic analysis of the language is not yet known? Without knowing if there is a morpheme boundary, a researcher might posit that two sounds are contrastive when really one is predictable with respect to the morphological structure. Alternatively, a researcher might assume that a longer constituent contains multiple morphemes in order to explain some phonological distribution, even without clear evidence this is true.

2.4 Predictability of distribution

The test of *predictability of distribution* relies on the distribution of sounds across a set of phonological (and, as discussed above, morphological) contexts. Two sounds A and B are traditionally considered to be contrastive if, in at least one phonological environment in

the language, it is impossible to predict which segment will occur. Distribution may be even more important for language acquisity than minimal pairs (Maye and Gerken 2000). Infants extract statistical regularities from the speech signal in order to acquire many types of distributional including acquire a language's segment inventory (Maye, Werker, & Gerken, 2002) and phonotactic structure (Chambers et al. 2003), and allophony (Peperkamp et al. 2006).

Because because the predictability of distribution is so robust, it may be a better test than lexical distinction (minimal pairs) for establishing phonemic contrast in polysynthetic languages. Distribution must be established across similar syllabic and morphological contexts (which may be difficult when the morphological analysis of the language is not known, as discussed above). It may also be useful to study a sound's phonological behavior by determining whether a sound patterns in a way that is similar to other elements in the system that are believed to be contrastive. This may require a study of existing descriptive and lexical resources, as in a corpus study, so that it is clear the patterns generalize across the lexicon and are not limited to cherry-picked datasets (see Andersson 2024 for an argument and case study of this methodology).

2.5 Learnability vs. economy

Finally, K.C. Hall (2013) points out that many of these tests point to conflicting or indeterminate analyses. This is especially true when two sounds are in an intermediate phonological relationship, where some instances are unpredictable and others are predictable. One option would be to apply the same analysis used for the predictable cases to the unpredictable cases. This is the approach often taken in generative phonology (e.g., Chomsky and Halle 1968) where a single phonological rule may account for alternations in derived environments as well as static distribution in non-derived environments. This approach is economical in the sense that it optimizes the phoneme inventory and number of features required, but it assumes that a learner can extend an analysis from alternations to unpredictable cases. An alternative view would be that learners cannot posit such abstract underlying forms for the unpredictable cases without

evidence of alternations. This is the approach taken under Lexicon Optimization (e.g., Prince and Smolensky 2002 [1993]: section 9; Tesar and Smolensky 1998; 2000).

The long mid vowels in Blackfoot can be analyzed as having an intermediate phonological relationship exactly of this kind, where some cases are unpredictable and others are predictable. While both of the approaches above are available for Blackfoot, I ultimately argue that the distributional evidence decides in favor of the second option, where the unpredictable long mid vowels do not have the same representation as the predictable long mid vowels. Before describing both types of long mid vowel, in the next section I turn to a brief overview of the language itself.

3 Language background

Blackfoot is the westernmost language of the Algonquian family, spoken in southern Alberta, Canada and northern Montana, USA (Mithun 1999:336–337; Frantz 2017:*xi*). It is the language of four distinct Indigenous Nations (three in Canada and one in the USA) that share a common language, culture, and heritage. Together, the four Nations form the Blackfoot Confederacy (Siksikaiitsitapi), which is an alliance of solidarity rather than a centralized governing body (Grinnell 1892:153; Juneau 2007:13*ff*; Dempsey 2019). Each Nation is associated with a distinct dialect, which are mutually intelligible, with small differences in phonology, morphology, and lexical items. See Weber et al. (2023) and Genee and Junker (2018) for more information about the dialects and speakers.

Blackfoot is a polysynthetic and strongly head-marking language (Bliss 2013). Like other Algonquian languages, words contain extreme phonological length and morphosyntactic complexity. Perhaps because of this, many aspects of the phonemic system are still debated (discussed in Author and Coauthor 2025). This is despite the fact that Blackfoot phonemic contrasts were largely settled in the 1970s (e.g., Frantz 1978), building on previous grammatical descriptions and transcription systems (e.g., Uhlenbeck 1938; Taylor 1969). My aim in this paper is to re-examine the diagnostics that were privileged in deciding what was phonemic or not and to bring further evidence to bear on the inventory contrasts. I focus on the vowel inventory and the segments allowed at the

left edge of roots. The observations in this paper are based primarily on written sources, as described below.

In the next section I introduce some aspects of Blackfoot grammar that are relevant for this paper. I first propose a modified vowel inventory and explain how it differs from previous proposals (Section 3.1). The next two sections describe syllable structure (Section 3.2.) and morphological structure (Section 3.3), because these both affect the distribution of long mid vowels. Finally, I briefly describe how I converted the practical orthography in Frantz and Russell (2017) to phonetic transcriptions (Section 3.4).

3.1 Phonemic inventory

The inventory of phonemic vowels which I argue for in this paper is given in Table 1. There are three short vowels /i o a/, and five long vowels /i: ε: a: ɔ: o:/.² The short mid vowels [ε] and [ɔ] are not phonemes, but do occur as allophones of long [ε:] and [ɔ:] before geminates and some consonant clusters, as I describe in Section 3.2. The actual pronunciation of /ε:/ and /ɔ:/ includes monophthong and diphthong allophones, depending on multiple factors, including phonological context, dialect, and speaker. I discuss this more in Section 4.3, where I defend the monophthongal representations.

Table 1: Blackfoot vowel phoneme inventory

	front	central	back
high	i i:		o o:
mid	ε:		ɔ:
low		a a:	

The inventory I propose here is novel, as most researchers assume that Blackfoot has a symmetrical six-vowel inventory which does not include /ε:/ or /ɔ:/ (e.g., Elfner 2006;

² The high back vowel varies between [o] and [u] (Author and Coauthor 2025:246). Algonquian studies typically use /o/ as the symbol for this vowel (e.g., Bloomfield 1946). As further evidence that this vowel is high, in Section 5.2 I show that long /o:/ patterns with long /i:/.

Frantz 2017). Other researchers do include short or long versions of the mid vowels [ɛ] and [ɔ] in the inventory (Taylor 1969:24, 32; Kinsella 1972:20; Van Der Mark 2003:8; Stacy 2004:10), although without supporting evidence. Earlier arguments that the long mid vowels are phonemic can be found in Weber (2020), Author and Coauthor (2024), and Author and Coauthor (2025) though none of these presents the same types or depth of evidence that I lay out here.

3.2 Syllable structure

Blackfoot distinguishes light syllables, CV, from heavy syllables, CVV and CVC, as in (1). These categories are relevant for vowel length neutralization and for the root alternations discussed later in the paper. The words in (1a) show that vowel length is distinctive in open syllables (CV vs. CVV). However, vowel length is neutralized to short in closed syllables, (1b) (only CVC allowed; *CVVC prohibited). This neutralization is also supported by morpheme alternations (e.g., the examples in Frantz 2017:2–3, which all include the durative prefix *á-* before the stem).

(1)	Shape	IPA	Orthography	Gloss	
a.	CV	[ʔâ:.ko.ka:]	<i>áakokaawa</i>	‘he will rope’	(BB)
	CVV	[ʔâ:.ko:.ka:]	<i>áakookaawa</i>	‘she will hold a Sundance’	(BB)
b.	CVC	[só.kaʔ.sim]	<i>sóka’simi</i>	‘shirt, dress’	(BB)
		[ʔóxʷ.ko.to.ki]	<i>óóhкотoki</i>	‘rock’	(BB)
		[pɛs.ká:n]	<i>passkaani</i>	‘dance’	(BB)
		[ʔim.mo.já:n]	<i>immoyaana</i>	‘fur coat’	(BB)
	CVVC	—	—		

The examples in (1) demonstrate a few other characteristics of Blackfoot phonotactics, described in greater detail in Author and Coauthor (2025). The only allowable codas, (1b), are a glottal stop [ʔ], a dorsal fricative [ç] ~ [x] ~ [xʷ], an alveolar fricative [s], or the first half of a geminate consonant, which I assume is doubly-linked to two syllables, following representations in Hayes (1989). The dorsal fricative is highly

restricted and only occurs before an obstruent. Vowel length neutralizes specifically to a centralized vowel [ɪ], [ɛ], [ɐ], [ɔ], or [ʊ] before /sC/ clusters and geminate consonants, but not before other coda consonants (Author and Coauthor 2025:250). A word-final consonant does not cause a long vowel to shorten, as in (1b) [ʔim.mo.já:n] ‘fur coat’, but a short vowel before a final consonant is always centralized, as in (1b) [só.kaʔ.sɪm] ‘shirt, dress’.

Onsets are required in Blackfoot and are always simplex. Consonants may be followed by a [j] glide, but Author and Coauthor (2025:258) argue that the [j] forms a rising diphthong with the following vowel. To fulfill the onset requirement, a glottal stop is epenthesized at the left edge of vowel-initial words, (2).

(2) UR	IPA	Orthography	Gloss	
/apí:t/	[ʔa.pí:t]	<i>apíít!</i>	‘sit!’	(BB)
/imitâ:/	[ʔi.mi.tâ:]	<i>imitááwa</i>	‘dog’	(BB)
/otán/	[ʔo.tén]	<i>otáni</i>	‘his/her daughter’	(BB)

Otherwise, vowel hiatus across morpheme boundaries is resolved via coalescence and other strategies (Elfner 2006:87ff; Bliss 2013:151–152; Weber 2020). Vowel coalescence frequently results in long [ɛ:] and [ɔ:], so I discuss this process thoroughly in Section 4.

3.3 Morphological structure

Morphological structure is relevant to this paper because there are predictable instances of the long mid vowels which result from vowel coalescence across morpheme boundaries. The basic morphological template for the verbal complex is given in (3) below (Taylor 1969; Armoskaite 2011; Bliss 2013; Frantz 2017; Déchaine and Weber 2018; Weber 2020). This template abstracts away from more complex details, and each position may itself be morphologically complex.

- (3) Morphological Template of the Verbal Complex
 person–(preverb*)–[initial–(medial)–final]–suffixes =clitics

The stem (in square brackets below) is itself morphologically complex and contains at least two elements, named ‘initial’ and ‘final’ for their position in the stem (Bloomfield 1946; Goddard 1990). The initial is minimally and prototypically a root, while the final is a verbalizing *v* head which derives stem classes based on transitivity and animacy (e.g., Bruening 2001:122; Brittain 2003; Hirose 2003; Branigan et al. 2005; Quinn 2006; Mathieu 2007; Slavin 2012). These may be separated by an optional nominal or classificatory root called a ‘medial’ (Dunham 2009; Biedny et al. 2021). The stem can be preceded by any number of optional prefixes known as “preverbs,” which have a wide variety of grammatical functions. In terms of inflection, a person prefix occurs in certain clause types, and the stem is followed by obligatory inflectional suffixes and DP enclitics (marking person and clause type), which differ for intransitive versus transitive verbs.

I illustrate this in (4) for intransitive verbs.³ The suffixes (bolded below) occur only in intransitive stems and agree with the grammatical (in)animacy of the single DP argument. In (4) the root *omahk-* ‘big’ combines with *-i* if the DP is animate, (4a), or *-o* if the DP is inanimate, (4b). The stem is given in square brackets.

(4) Blackfoot intransitive verbs

- | | |
|--|---|
| a. <i>áakomahksimma</i>
áak-[omahk- i]-mm-a
FUT-[large- AI]-IND-3
‘it (e.g., the pot) will be large’ [FR 190] | b. <i>áakomahk^owa</i>
áak-[omahk- o]-wa
FUT-[large- II]-3
‘it will be big’ (BB) |
|--|---|

³ Abbreviations are as follows: 1 = first person, 2 = second person, 3 = third person, 21 = first person inclusive, AI = animate intransitive, AN = animate, BB = Tootsinám (Beatrice Bullshields), B = Blackfoot, C = consonant, CAUS = causative, DP = determiner phrase, FR = Frantz and Russell (2017), FUT = future, G = glide, GET = get, acquire, IC = initial change, II = inanimate intransitive, IMP = imperative, IN = inanimate, IND = indicative, INS = instrumental, IPA = International Phonetic Alphabet, IPFV = imperfective, LOC = locative, NEG = negation, NONAFF = nonaffirmative, OBJ = object, OBV = obviative, PA = Proto-Algonquian, PL = plural, PRF = perfect, REFL = reflexive, SG = singular, STICK = wood classifier, SUB = subject, TA = transitive animate, TI = transitive inanimate, UR = underlying (phonological) representation, V = vowel. Additional symbols include: *v* = syntactic head which introduces an external argument, σ = syllable, μ = mora, [] = phonetic transcription of speech and morphological stem (in interlinear glosses), [] = phonetic transcription based on orthography.

The root alternations later in the paper focus on the boundary at the left edge of the stem (e.g., word-initially when there are no prefixes, or between a person prefix or preverb and the initial).

3.4 Orthography and transcription

Unless otherwise noted, the orthographic transcriptions in this paper come from the most recent dictionary of Blackfoot (Frantz and Russell 2017; a digitized version of this dictionary has been augmented with additional examples and media files and is described in Genée and Junker 2018). Examples use a five line gloss. The first line contains a phonetic transcription that is derived from the orthographic transcription on the second line. The third and fourth lines contain a morphemic analysis, and the final line includes the translation. I only show the verb in this paper, even if it was part of a multi-word example in an entry in Frantz and Russell (2017).

I use double square brackets around phonetic transcriptions which are derived from the orthography. These transcriptions were created following rules discussed in Frantz (1978, 2017) and recent phonological descriptions of the language (Elfner 2006; Denzer-King 2009, 2012; Goad and Shimada 2014a,b; Weber 2020; Author and Coauthor 2025). In particular, syllabification follows the generalizations in Author and Coauthor (2025). Free translations are verbatim from the dictionary, except when the word was extracted from a longer example sentence, in which case I modified the translation accordingly.

I use [ɛ:] and [ɔ:] to represent the spellings <ai> and <ao> (in Frantz's orthography) in open syllables. I use short [ɛ] and [ɔ] in closed syllables, following the rule of vowel shortening discussed in Section 3.2. It is important to note that my transcriptions of these vowels are necessarily broad. Previous research clearly shows that there are variant pronunciations of both sounds (Frantz 1978, 2017:2–3, 183; Taylor 1969:86–88; Kaneko 1999; Peter 2014), but researchers do not agree on the variants or their contexts. See Author and Coauthor (2024) and Section 4.3 for more explanation.

In the next section I discuss unpredictable and predictable instances of the long mid vowels [ɛ:] and [ɔ:]. It is important to establish the full set of phonemic vowels because in later sections I turn to morphophonological alternations of roots that begin in vowels.

4 Long mid vowels are phonemic

There are two types of long mid vowels in Blackfoot: predictable and unpredictable. Predictable long mid vowels arise from vowel coalescence across a morpheme boundary (Section 4.1). A small number of long mid vowels occur unpredictably within morphemes, where they do not participate in any alternations (Section 4.2). After laying out the evidence for each type, in Section 4.3 I argue against an alternative analysis that treats [ɛ:] and [ɔ:] as allophones of underlying diphthongs /ai/ and /ao/. In Section 4.4 I conclude that some long mid vowels /ɛ:/ and /ɔ:/ are phonemic in Blackfoot, necessitating the inventory from Section 3.1 with five long vowels.

4.1 Predictable long mid vowels: the result of coalescence

Most surface instances of long mid vowels in Blackfoot derive from underlying vowel sequences across morpheme boundaries (Uhlenbeck 1938:4; Taylor 1969:86–93; Kinsella 1972:20; Elfner 2006:96ff; Frantz 2017:186; Weber 2020:129–130, 238ff; Author and Coauthor ita 2024:272–273; Weber et al. 2023:1221–1222). Specifically, an underlying sequence /a+i/ resolves to [ɛ:] (in open syllables) and /a+o/ resolves to [ɔ:] (in open syllables). This is demonstrated in (5) and (6) below at a stem-internal boundary.⁴

⁴ Previous studies of Blackfoot demonstrate coalescence between the stem and inflectional suffixes (Frantz 2017:8–10; Windsor 2017:7), across the left edge of the stem (Bliss 2013:151; Frantz 2017:2–3; Windsor 2017:7), or in a range of positions (Elfner 2006:98). The reason I use a stem-internal boundary is that vowel hiatus in Blackfoot has been argued to resolve differently at the left edge of the stem: Windsor (2017) found that coalescence is optional across the left edge of the stem, but obligatory within stems; and Weber (2020:177ff, 237ff) shows that some vowel combinations resolve via different processes across the left edge of the stem versus stem-internally. The left edge of the stem is a prosodically strong boundary in many Algonquian languages and often has different sandhi processes than within the stem (e.g., Bloomfield 1946; Goddard 1990; Author et al. 2025).

The transitive animate (TA) suffix *-istoto* ‘CAUS’ begins in a vowel [i], which is apparent after the consonant-final root *sap-* ‘correct’ in (5a).⁵ This vowel coalesces with a preceding [a], such as the [a] at the right edge of an initial like *sata-* ‘offended’, (5b). The pronunciation of the underlying sequence /a+i/ is represented as [ɛ:] in (5b) below (discussed further in Section 4.3). This example is also discussed in Author and Coauthor (2024:273), though they represent the coalesced vowel as [æ:], following Frantz’s (2017:2–3) description of the Káínai pronunciation.

(5) Underlying /a+i/ vowel sequences

- | | |
|--|---|
| a. [[sa.pí. ^s to.tó:.sə]]
<i>sapístotoósa</i>
[sap–istoto]–:s
[correct–CAUS.TA]–2SG:3.IMP
‘reach an agreement with him!’ [FR 242] | b. [[sa.té:. ^s to.to:.sə]]
<i>satáístotoosa</i>
[sata–istoto]–:s
[offended–CAUS.TA]–2SG:3.IMP
‘purposely make her angry!’ [FR 244] |
|--|---|

The animate intransitive (AI) suffix *-opii* ‘sit’ begins in a vowel [o] after the consonant-final root *sap-* ‘in’ in (6a). This vowel coalesces with a preceding [a], such as the [a] at the right edge of an initial like *ipakkssa-* ‘bare’, (6b). The pronunciation of the underlying sequence /a+o/ is represented [ɔ:] below (discussed further in Section 4.3).

(6) Underlying /a+o/ vowel sequences

- | | |
|--|--|
| a. [[ni.tâ:.ks.sa.pó.pi]]
<i>nitáakssapopii</i>
nit–aak–[sap–opii]
1–FUT–[in–sit.AI]
‘I’ll ride in (a vehicle)’ [FR 243] | b. [[ʔi.pək.ks.sɔ:.pi:.wə]]
<i>ipakkssaopiiwa</i>
[ipakkssa–opii]–wa
[bare–sit.AI]–3
‘he’s sitting with nothing on
(in the nude)’ [FR 81] |
|--|--|

In summary, the underlying sequence /a+i/ resolves to [ɛ:] within the verb stem, while /a+o/ resolves to [ɔ:]. The underlying vowel qualities are revealed next to

⁵ The morphemic analysis of Blackfoot verb stems is not always explicitly argued for (see Weber 2020, 2021). Simply looking at an example like (5b) does not tell us whether the suffix begins in [i], [ɛ:], or something else. I always include an example of the suffix after a consonant, like (5a), in order to determine the underlying vowel quality at the left edge of the suffix.

consonants, and the surface instances of [ɛ:] and [ɔ:] are predictable realizations of the underlying vowel sequences. (See Elfner 2006 and Weber 2020 for discussion of other vowel hiatus resolution strategies.) The vast majority of surface [ɛ:] and [ɔ:] arise in these contexts. In the next section I turn to a discussion of unpredictable long mid vowels.

4.2 Contrastive long mid vowels: lexical distinctiveness and overlapping distribution

Some instances of [ɛ:] and [ɔ:] are tautomorphemic, where they are not predictable elements of the word and they represent abstract units of contrast. The evidence comes from tests of *lexical distinctiveness* (minimal pairs) and *overlapping distribution* with other long vowels, both of which are standard tests for determining phonemic contrast (Steriade 2007; Dresher 2009:72; Dresher 2011; D.C. Hall 2011; K.C. Hall 2013).

There is a minimal pair for [ɔ:] vs. [a:] in (7), which shows that [ɔ:] and [a:] can distinguish lexical items.⁶ This minimal pair was confirmed with Tootsinám (Beatrice Bullshields), a speaker of Káínai Blackfoot.

- | | |
|--|------------------------|
| (7) Underlying tautomorphemic [ɔ:] in contrast with [a:] | |
| a. [[ʔɔ:.ní:t]] | b. [[ʔa:.ní:t]] |
| <i>aoníít</i> | <i>aaníít</i> |
| [ao -nii]-t | [aan -ii]-t |
| [hole-by.needle.TI]-2SG.IMP | [say-AI]-2SG.IMP |
| ‘pierce it!’ [FR 301] | ‘say (s.t.)!’ [FR 289] |

As I discussed in the previous section, many long mid vowels are the result of vowel coalescence across a morpheme boundary. Therefore it is important to use derivational paradigms (Bauer 1997) to show that the the mid vowel [ɔ:] in (7a) is truly tautomorphemic and that the morphemic analysis in (7a) is correct. The root *ao*- ‘hole’ is an autonomous morpheme which can occur before other suffixes, such as the medial *-ks* ‘STICK’ in (8).

⁶ Further examples of tautomorphemic, unpredictable [ɛ:] and [ɔ:] that do not form minimal pairs are included in Weber (2020:Appendix A).

390

- 391 (8) [ʔɔ:.ks.tsít]
 392 *aokstsít*
 393 [aɔ-ks-htsi]-t
 394 [hole-STICK-by.mouth.TI]-2SG.IMP
 395 ‘gnaw a hole in it!’ [FR 301]
 396

397 I know of no (near) minimal pairs between [ɛ:] and other long vowels, but [ɛ:] in (9a)
 398 occurs at the beginning of the word after an epenthetic [ʔ], just like [ɔ:] and [a:] in (7). As
 399 (9b) shows, the long [ɛ:] occurs internal to a morpheme *aist-* ‘here’, and therefore does
 400 not result from vowel coalescence.

401

- 402 (9) a. [ʔɛ:.stó:t] b. [ʔɛ:.stá:.pi.ksít]
 403 *aistóót!* *aistáápiksit!*
 404 [aist-oo]-t [aist-aapiksi]-t
 405 [here-go.AI]-2SG.IMP [here-throw.IN]-2SG.IMP
 406 ‘come here!’ [FR 300] ‘throw it here!’ [FR 291]
 407

408 The exhaustive list of vowels that can occur in a word-initial frame [ʔ__...] include:
 409 three short vowel qualities, (10), and five long vowel qualities, (11).⁷ Again, this is a
 410 distributional argument for contrast, and it is exactly what we expect if these eight vowels
 411 form the phonemic inventory. Further examples of long [ɛ:] and [ɔ:] in the same local
 412 context as other long vowels are given in Weber (2020:42–43), Author and Coauthor
 413 (2024:274), and Author and Coauthor (2025:244–245).

414

- 415 (10) Short vowels at the left edge of the word
 416 *IPA Orthography Gloss*
 417 [i] [ʔi.tsín.ʔx^w.to:t] *itsínohtoot!* ‘place it among the rest!’ FR 120
 418 [o] [ʔo.ká:t] *okáát!* ‘rope!’ FR 182
 419 [a] [ʔa.ks.ta.kít] *akstakít!* ‘read!’ FR 188
 420

⁷ There is also exactly one stem in Frantz & Russell (2017) which is listed with an initial <oi> digraph, representing an initial diphthong [oi]. As I discuss in Section 4.3.2, this [oi] is not tautomorphemic, but derived, and is apparently a variant pronunciation of <o>.

421	(11)	Long vowels at the left edge of the word			
422		<i>IPA</i>	<i>Orthography</i>	<i>Gloss</i>	
423	[i:]	[[ʔi:.ts:.ká:t]]	<i>ĩtsskáát!</i>	‘fight!’	FR 38
424	[o:]	[[ʔô:.ká.ta.kɪt]]	<i>ókátakit!</i>	‘bead!’	FR 201
425	[a:]	[[ʔa:.kx ^w .kí.ma:t]]	<i>aakohkímaat!</i>	‘argue!’	FR 285
426	[ɛ:]	[[ʔɛ:. ^s tó:t]]	<i>aístóót!</i>	‘come here!’	FR 300
427	[ɔ:]	[[ʔɔ:.ní:t]]	<i>aoníít!</i>	‘pierce it!’	FR 301
428					

429 Taken together, the examples above show that long mid vowels meet the two main
430 criteria for contrast (K.C. Hall 2013): *lexical distinctiveness* for [ɔ:], and *overlapping*
431 *distribution* for both [ɛ:] and [ɔ:]. I conclude that /ɛ:/ and /ɔ:/ are phonemic, at least in the
432 words above.

433 4.3 Against underlying diphthongs

434 The description and analysis thus far assumes that the long mid vowels are underlying
435 monophthongs, /ɛ:/ and /ɔ:/. There is an obvious alternative analysis – argued for
436 explicitly in Frantz (2017:186) – that treats these as underlying diphthongs or vowel
437 sequences instead, /ai/ and /ao/. In this section I argue against such a diphthong analysis.
438 The monophthong analysis is supported by allophonic and distributional arguments.

439 4.3.1 Allophony and phonetic variation

440 Although the long mid vowels have monophthong and diphthong positional variants, the
441 monophthongs occur in the least restricted contexts and are the most parsimonious
442 choices for phonemic representations. According to Frantz (1978:312; 2017:2–3), the
443 back vowel is a diphthong [aw] before a glottal stop, a short monophthong [ɔ] before long
444 consonants (i.e., in closed syllables, as discussed in Section 3.2), and a long
445 monophthong [ɔ:] elsewhere. I therefore take /ɔ:/ to be the phonemic representation of
446 this sound. Frantz (1978:312; 2017:2–3) describes the front vowel in terms of the same
447 phonological positions, but with dialectal differences as well. For the two dialects he
448 considers, both dialects have a diphthong before a glottal stop ([ej] for Káínai and [aj] for
449 Siksiká), and both dialects have a short monophthong [ɛ] before long consonants (i.e., in
450 closed syllables, as discussed in Section 3.2). Elsewhere, Káínai uses a monophthong [æ:]
451 and Siksiká uses a diphthong [ej]. If the underlying phoneme matched the elsewhere

allophone it would be /æ:/ in Káínai and /ej/ in Siksiká, making Siksiká the only dialect with a single phonemic diphthong. For this reason, I assume the underlying phoneme is a monophthong in all dialects.

I use /ɛ:/ for the front vowel to create a symmetrical vowel system, but also because [ɛ(:)] is the most frequently used vowel transcription in other Blackfoot descriptions (Uhlenbeck 1938; Taylor 1969; Kinsella 1972; Frantz 1978; Kaneko 1999; Van Der Mark 2003; Stacy 2004; Elfner 2006; Derrick 2007; Denzer-King 2009; Peter 2014; Windsor 2017; Weber 2020). A recent articulatory description (Author and Coauthor 2025) also supports /ɛ:/ for the front vowel. The acoustic findings in this paper are based on a dataset of 177 words, each repeated one to five times by Tootsinám (Beatrice Bullshields), a female speaker of Káínai Blackfoot. The figure and caption below are repeated from Author and Coauthor (2025:246). This figure shows the F1 and F2 values for all eight phonemic vowels based on 1,089 vowel tokens. As shown here, the mid vowels are midway in height between the high and low vowels. This makes the front vowel approximately equivalent to [ɛ:] for this speaker rather than the [æ:] that Frantz (2017:2–3) describes.

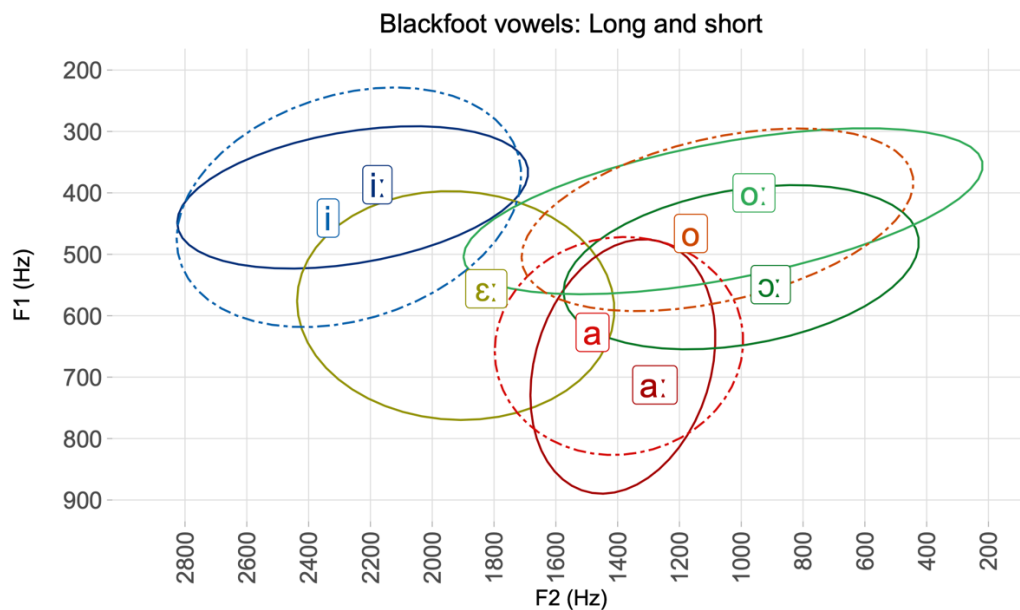


Figure 1. Phonemic long and short Blackfoot vowels (based on 1089 vowel tokens: [i] = 303, [i:] = 144, [ɛ:] 65, [a] = 157, [a:] 184, [ɔ:] = 32, o = 145, [o:] = 59; 1.5 standard deviation outlines shown for clarity). Solid outlines are phonemic long vowels and dashed/dotted lines are phonemic short vowels.

Finally, there is clearly a lot of speaker variation in how these vowels are pronounced. Several descriptions include slightly different allophones and environments than Frantz (1978:312; 2017:2–3, 183) does (e.g., Taylor 1969:86–88; Lowery 1979:75; and more recently in Peter 2014:23). Some research with narrow transcriptions also shows there is intraspeaker variation (e.g., Kaneko 1999; Peter 2014; Author and Coauthor 2024). Author and Coauthor (2024) is a preliminary corpus study of 149 tokens of the Blackfoot digraphs <ai>, <ao>, and <oi> extracted from short stories told by three speakers, each representing a different dialect (Káínai, Aapátóhsipikani, and Aamsskáápipikani). The Siksiká dialect, which uses a diphthong for the elsewhere allophone, is not represented. This study found that each speaker used a different ratio of monophthongs to diphthongs across environments and even in the “elsewhere” environments, but that monophthongs were more frequent overall. They also found that both types of pronunciation occurred before glottal stops, contrary to the description in Frantz (2017:2–3). The extent of speaker variation I leave for future studies, but it is important to note here that no study suggests that diphthongs are the context-free allophone.

4.3.2 Distributional restrictions

The diphthong analysis treats all surface instances of [ɛ:] and [ɔ:] as one pronunciation of underlying diphthongs, /ai/ and /ao/. The idea is that learners could extend this analysis from the derived cases across morpheme boundaries to the (infrequent) morpheme-internal cases on grounds of economy: it optimizes the phoneme inventory and number of features required. As mentioned above, this analysis is explicitly adopted in Frantz (2017:186). However, there is another diphthong, /oi/, which arises across morpheme boundaries and only across morpheme boundaries. It does not have the same distribution as [ɛ:] and [ɔ:] and does not contrast with other long vowels, which would be unexpected

if all three vowels were underlying diphthongs, /ai/, /ao/, and /oi/. Those distributional restrictions are easily explained if [ɛ:] and [ɔ:] have a different representation, such as the monophthongs /ɛ:/ and /ɔ:/.

An example of derived /o+i/ is in (12). The animate intransitive (AI) final *-ipoyi* ‘stand’ begins in a vowel [i] after a consonant-final initial, (12a). This vowel diphthongizes with a preceding [o], such as the [o] at the right edge of an initial like *ikssto-* ‘abreast’, (12b). The pronunciation of the underlying sequence /o+i/ is represented as [oi] in (12b) below.

(12) Underlying /o+i/ vowel sequences

- | | |
|--|--|
| <p>a. [[oʔtsipójit]]
 <i>oʔtsipóyit!</i>
 [oʔt-<i>ipoyi</i>]-t
 [here-stand.AI]-2SG.IMP
 ‘stand by me!’ [FR 220]</p> | <p>b. [[ó:paksíks:toipojja:wə]]
 <i>áópaksíksstoipoyiyaawa</i>
 a-opak-[ikssto-<i>ipoyi</i>]-yi=aawa
 IPFV-flat-[abreast-stand.AI]-3PL=PRX.PL
 ‘they stand abreast’ [FR 204]</p> |
|--|--|

Unlike [ɛ:] and [ɔ:], this diphthong is only pronounced as a monophthong [ɤ] or [ɥ] in closed syllables by a subset of speakers (Taylor 1969:32–33; Elfner 2006:109; Frantz 2017:3). The diphthong /oi/ also does not have the same distribution as [ɛ:] and [ɔ:]. First, it never occurs after a word-initial glottal stop like the long vowels in (11).⁸ Second, I have found no examples of tautomorphemic [oi], even with the help of a lexical database which includes morphosyntactic structure (described in Section 8.3). If underlying vowel sequences like /ai/ and /ao/ can occur in these contexts, then it is unclear why /oi/ never does. If instead [ɛ:] and [ɔ:] are underlying monophthongs then the distributional differences are easily explained. One final distributional argument is discussed in Section 5.2.3: a root-initial /j/ may be followed by long [i:], [a:], and [o:], but not [ɛ:] or [ɔ:]. If

⁸ Frantz & Russell (2017:182) lists a single stem that begins with <oi>, as in *óissimisa!* ‘regard him as your son-in-law!’ Unlike the word-initial vowels in (11), this is a derived diphthong: it involves a third person prefix, *o-*, followed by a noun stem, *-iss* ‘son in law’, followed by a verbalizing final, *-imm* TA ‘by mind’. It is also apparently a variant pronunciation of a monophthongal <o>, which occurs in the possessed noun *óssiksi* ‘his/her sons-in-law’ (Frantz & Russell 2017:100).

[ɛ:] and [ɔ:] are underlying /a/-initial diphthongs, it unclear why long [a:] is allowed here, but not /ai/ and /ao/.⁹

4.4 Summary

To summarize, long mid vowels in Blackfoot divide into two types. First, there is a large set which are predictable from underlying vowel sequences across morpheme boundaries. Second, there is a relatively small set of long mid vowels which are unpredictable, morpheme-internal, and contrast with the other long vowels. I argued that these vowels are phonemic long monophthongs, and that an alternative analysis which treats them as underlying vowel sequences fails to explain their allophony and distribution. The next sections present evidence for the phonemic status of /ɛ:/ and /ɔ:/ specifically at the left edge of roots based on patterns of alternation (Section 5) and a small corpus study (Section 6).

5 Root-initial labiovelar glides are epenthetic

In the accepted phonemic analysis of Blackfoot, no roots begin in an underlying non-high long vowel. Instead, roots that have a long vowel at the left edge of the word after an epenthetic [ʔ] are argued to begin in an glide /w/ which then deletes at the left edge of the word (Frantz 2017; Frantz and Russell 2017). Under this analysis, *no* roots begin with underlying /ɛ:/ or /ɔ:/, which would be an odd lexical gap if these two vowels truly are phonemic. In this section I describe the morphophonological alternations of roots that surface with a long vowel word-initially. I show that many of these roots have variants after a vowel that begin in either a [w] or a [j]. However, there are key differences between these two groups that lead me to argue that the [w] at the left edge of roots is epenthetic while the /j/ is underlying but deletes word-initially. Under this analysis, there

⁹ Interestingly, the diphthong [oi] occurs after a root-initial [j] in two stems in Frantz & Russell (2017:318) beginning with the root *yoiisst-* ‘tough, sinewy’. These are not obviously derived, but like the case in footnote 10 they are apparently a variant pronunciation of monophthongal <o>. One of the two stems is listed again with the root spelled as *yosst-* (Frantz & Russell 2017:320), and both are related to the noun stem *mósstsi-* NA ‘vein’ or *osstsi-* NA ‘blood vessel’ (Frantz & Russell 2017:153, 207).

are five patterns of alternation for roots which begin in underlying long vowels, consistent with an inventory that includes five long vowels. This means that roots can indeed begin in underlying long mid vowels, as expected if those vowels are phonemic.

The two relevant positions are when the root stands at the left edge of the verbal complex versus after a prefix (ending in a consonant or a vowel). The “left edge” has to be carefully defined. Some morphological orders use no prefixes for person agreement. An example is the imperative in (13a), where the root stands at the left edge. In the indicative order, a root is never at the left edge in verbs inflected for first or second person arguments, which are indexed on the verb with a prefix. Although the indicative order also generally uses no prefixes for third, non-stative verbs with no preverbs undergo a process of morphological ablaut known in Algonquianist studies as “initial change” (Costa 1996).¹⁰ The indicative verb in (13b) has a long high front vowel, as opposed to the short low vowel in the imperative verb in (13a).¹¹ This process of initial change neutralizes contrasts at the left edge of the verbal complex in the indicative order. For that reason, I only consider a root to be at the left edge if it occurs at the left of an unprefix noun, an unprefix stative verb, or an unprefix imperative.

(13) Non-stative verbs

a. *Imperative*

[[ʔa.tsi.ni.kít]]

atsinikit

[atsinik-i]-t

[relate.story-AI]-2SG.IMP

‘relate a story!’ [FR 120]

b. *Indicative*

[[ʔi:.tsi.ní.ki.wə]]

ütsinikiwa

[i\atsinik-i]-wa

[IC\relate.story-AI]-3

‘he told a story’ [FR 120]

In the next section, I show that glides are prohibited at the left edge. After that I turn to a discussion of alternations of roots that begin in long vowels at the left edge.

¹⁰ Initial change in Blackfoot has been more narrowly defined to ablaut which affects a vowel after a consonant (Taylor 1967, 1969:120ff). The differences between (13a,b) are instead described as roots with two different shapes in complementary distribution across ‘initial’ positions (Taylor 1969:128ff).

¹¹ This word has pitch accent on the fourth (final) syllable in the dictionary, but in my own fieldwork is always pronounced with pitch accent on the third syllable.

5.1 Restrictions on glides

Glides never occur at the left edge of the word (Frantz 2017:177, Rule 15; Weber 2020). The words in (14) demonstrate that all short consonants except for glides can occur at the left edge of the word. The sound /x/ is not included because it only occurs in a coda position. Word-initial [ʔ] is not contrastive, and is predictably epenthesized before word-initial vowels. Although [j] can form a rising diphthong with a following vowel (Author and Coauthor 2025:258), this is apparently only possible after a supralaryngeal consonant, because sequences like [ʔjV...] never occur.

(14) Consonants at the left edge of the word

[p]	[[pí:.ta:.wə]]	<i>píitaawa</i>	‘eagle’
[t]	[[tá.tsi.kík.kó.na.ma:.nə]]	<i>tátsikikkónamaana</i>	‘center pole of Sundance tent’
[k]	[[kó:.pi.sí]]	<i>kóópisi</i>	‘soup’
[s]	[[sɾ̥t.tsí.wə]]	<i>sisttsíwa</i>	‘small bird’
[ts]	[[tsi.má]]	<i>tsimá</i>	‘where’
[ks]	[[ksi.só.ji.wə]]	<i>ksisójiwa</i>	‘tea kettle’
[m]	[[ma.mí:.wə]]	<i>mamíiwa</i>	‘fish’
[n]	[[na.tá.jo.wə]]	<i>natáyowa</i>	‘lynx’
*[w]	—	—	
*[j]	—	—	
[ʔ]	[[ʔa.tsi.kí.ní]]	<i>atsikíni</i>	‘shoe’

Glides can otherwise occur as onsets to word-medial syllables. This is demonstrated in (15), where all short consonants including glides occur between vowels. The glottal stop only occurs rarely between vowels. It is not epenthetic in this position, as some words allow vowels in hiatus. For example, compare [[na:.áx.sə]] *naaáhsa* ‘my grandparent’ with no glottal stop between the vowels to the vocative [[na.ʔá]] *na ’á* ‘mother!’ in (15).

604	(15)	Intervocalic consonants as onsets			
605	[p]	[[si.no.pá:.wə]]	<i>sinopááwa</i>	‘swift fox’	
606	[t]	[[ma.tó:.miç.ka:t]]	<i>matóómiihkaat</i>	‘go fishing!’	
607	[k]	[[ʔi st .tsi.kí:.wə]]	<i>isttsikííwa</i>	‘it is slippery’	
608	[s]	[[ʔa.so.jín.ni]]	<i>asoyínni</i>	‘barrel’	
609	[ts]	[[ʔa.tsi.kí.ni]]	<i>atsikíni</i>	‘shoe’	
610	[ks]	[[ʔi.sa.ksí.wə]]	<i>isaksíwa</i>	‘he went out’	
611	[m]	[[ʔâ:.ko.mɛs.sa.pi.wə]]	<i>áakomaissapiwa</i>	‘s/he will stare’	
612	[n]	[[ní.na:.wə]]	<i>nínaawa</i>	‘chief, man’	
613	[w]	[[ʔa.wóʔ.ta:.ni]]	<i>awó’taani</i>	‘shield’	
614	[j]	[[ʔá.joʔ.ka:.wə]]	<i>áyo’kaawa</i>	‘she is sleeping’	
615	[ʔ]	[[na.ʔá]]	<i>na’á</i>	‘mother!’ (voc.)	

616
617 Table 2 below summarizes the distribution of onsets in (14) and (15). The shaded
618 cells in the table illustrate a gap: all of the short consonants which can occur as word-
619 medial onsets can also occur at the left edge, *except* for glides.

620

621 **Table 2:** Allowable onsets at the left edge of the verbal complex versus between vowels

	p	t	k	s	ts	ks	m	n	w	j	ʔ
Left edge	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✓
Intervocalic	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

622

623 Elfner (2006:91) and Bliss (2013:152) argue that underlying /w/ and /j/ delete at the
624 left edge. However, I argue in the following sections that the distributions of [w] and [j]
625 at the left edge of roots are different. I find evidence that the [w] is actually epenthetic in
626 certain positions, while /j/ is underlying and deletes at the left edge. All roots with
627 epenthetic [w] can be reanalyzed as beginning in an underlying long vowel. As a
628 consequence, roots begin underlyingly in all five long vowels, including the mid vowels,
629 which is the expected pattern if /ɛ:/ and /ɔ:/ are phonemic.

5.2 Alternations: Roots beginning in long vowels at the left edge

To preview the findings in this section, roots that begin in long vowels at the left edge participate in one of three types of alternation, schematized in (16). The first group begins in a long vowel also after a prefix, (16a). If the prefix ends in a vowel, then that vowel coalesces with the first vowel of the root. The second group begins in a glide [w] after a prefix, (16b). The patterns in (16a) and (16b) occur for complementary sets of long vowels. The third group begins in a glide [j] after a prefix, (16c).

(16)	Left edge	After prefix	Vowel qualities
a.	V:...	V:...	high vowels {i:, o:}
b.	V:...	wV:...	non-high {ɛ:, a:, ɔ:}
c.	V:...	jV:...	{i:, a:, o:}

In the next subsections I present data for all three groups, and following that I argue for underlying representations for all three groups. The examples are restricted to roots which begin with a long vowel that is parsed into an open syllable. The same three types of alternations also occur for roots which begin in a closed syllable. See Section 6 for corpus data that confirms this, or Weber (2020:122ff) for examples of roots that begin in closed syllables participating in these alternations.

5.2.1 Long vowels after a prefix

For the first group of roots, the vowel at the left edge of the verbal complex is either a long [i:] or long [o:], and the root allomorph after a prefix also begins with long [i:] or [o:], which coalesces with a preceding vowel. For example, *iitssk-* ‘scuffle’ begins with a long [i:] at the left edge, (17a), after a consonant (17b), and after a vowel, (17c), although in this position the long [i:] coalesces with the preceding [a] to form [ɛ:i].¹² Similarly,

¹² The IPA transcription in these two examples follows the orthography as well as the pronunciation of Tootsinám (Beatrice Bullshields) from my own fieldwork with her. She sometimes pronounces the coalesced vowels in (17) and (18) as a simple [ɛ:] or [ɔ:] in fast speech, and I would not be surprised if this

655 *ook-* ‘scuffle’ begins with a long [o:] in all three positions, (18), though the long [o:]
 656 coalesces with a preceding [a] in to form [ɔ:o], (18c).

657

658 (17) a. Left edge
 659 [[ʔi:.ts:.ká:t]]
 660 *íitsskáát*
 661 [íitssk-aa]-t
 662 [scuffle-AI]-2SG.IMP
 663 ‘fight!’ [FR 38]

665 b. After C
 666 [[ʔâ:.ki:.ts:.ka:.wə]]
 667 *áakíitsskaawa*
 668 aak-[íitssk-aa]-wa
 669 FUT-[scuffle-AI]-3
 670 ‘she will fight’ [FR 38]

c. After V
 [[ʔé:.so:.ké:i.ts:.ka:.wə]]
áísookáíitsskaawa
 a-isooka-[íitssk-aa]-wa
 IPFV-used.to-[scuffle-AI]-3
 ‘he used to fight’ [FR319]

672 (18) a. Left edge
 673 [[ʔô:.ká.ta.kít]]
 674 *ókátakit*
 675 [ook-ataki]-t
 676 [bead-AI]-2SG.IMP
 677 ‘bead!’ [FR 201]

679 b. After C
 680 [[ʔâ:.ko:ka.ta.ki.wə]]
 681 *áakookátakit*
 682 aak-[ook-ataki]-wa
 683 FUT-[bead-AI]-3
 684 ‘bead!’ [FR 201]

c. After V
 [[ʔó:ô.ka.ta.ki.wə]]
áóokatakiwa
 a-[ook-ataki]-wa
 IPFV-[bead-AI]-3
 ‘she does beadwork’ [FR 201]

686 Abstracting away from vowel coalescence and the predictable [ʔ] that occurs at the
 687 left edge of the verbal complex, the first set of roots begins in a long high vowel at the
 688 left edge and also after a prefix ending in a consonant or vowel, (19a,b).

689

690 (19) Left edge After C After V Gloss

was a common speaker variant. I leave this type of coalesced vowel for future research, though see Weber (2020:146ff) for one analysis.

691	a.	[[ʔi:ts:k-]]	[[i:ts:k-]]	[[i:ts:k-]]	‘scuffle’
692	b.	[[ʔo:k-]]	[[o:k-]]	[[o:k-]]	‘bead’
693					

694 5.2.2 [w] glide after a prefix

695 For the second group of roots, the root begins with long [a:], [ɛ:], or [ɔ:] at the left edge, a
696 short vowel after a consonant, and [w] following [a], (20)–(22). Frantz (2017:86) calls
697 these ‘variable-length vowels’, and the length of the vowel [a:] ~ [a] is reflected in the
698 orthography. For example, *aan-* ‘tell’ begins with a long [a:] at the left edge, (20a), a
699 short [a] after a consonant, (20b), or [wa:] after the vowel [a], (20c).

700

701 (20) a. Left edge

702 [[ʔa:.ní.ʰtsís]]
703 *aanistsísa*
704 [aan-ist]-:s
705 [tell-TA]-2SG:3.IMP
706 ‘tell her!’ [FR 289]

707

708 b. After C

709 [[sa.pa.ní.ʰtsís]]
710 *sapanistsísa*
711 sap-[aan-ist]-:s
712 correct-[tell-TA]-2SG:3.IMP
713 ‘respond correctly to him!’ [FR 241]

714

708 c. After V

[[ʔi.ts:.tsá:.wa:.ní.ʰtsís]]
itsstsááwaanistsísa
itsstsaa-[aan-ist]-:s
callous-[tell-TA]-2SG:3.IMP
‘be blunt, callous with her!’ [FR 123]

715 The pattern in (21) and (22) for roots that begin in [ɛ:] and [ɔ:] is similar: these roots
716 also begin with a long vowel at the left edge in (21a) and (22a), or a [w] after the vowel
717 [a] in (21c) and (22c). The length of the vowel after a consonant in (21b) and (22b) is
718 indeterminate, because the vowels [ɛ:] and [ɔ:] are always written with the digraphs <ao>
719 and <ai>, even in phonological contexts where these vowels are predictably short (e.g.,
720 before geminates; see Section 3.2). A targeted acoustic study could determine whether
721 [ɛ:] and [ɔ:] pattern with [a:] and shorten in this context; if they do, this would be
722 additional evidence that all three non-high long vowels pattern as a class. Even without
723 this acoustic evidence, it is clear that [a:], [ɛ:], and [ɔ:] all share a Ø ~ [w] alternation,
724 described below.

725

726	(21) a. Left edge			
727		[[ʔɛ:. ^s tó:t]]		
728		<i>aístóót</i>		
729		[aíst-oo]-t		
730		[towards-go.AI]-2SG.IMP		
731		‘come here!’ [FR 300]		
732				
733	b. After C		c. After V	
734		[[nox ^w .ké(:). ^s to:t]]		[[ʔá.wɛ:. ^s tɛç.tsi:.wə]]
735		<i>noohkaístóót</i>		<i>áwaístaihtsiwa</i>
736		noohk-[aíst-oo]-t		a-[aíst-aihtii]-wa
737		please-[towards-go.AI]-2SG.IMP		IPFV-[towards-move.II]-3
738		‘please come here!’ [FR 162]		‘it is coming’ [FR 300]
739				
740	(22) a. Left edge			
741		[[ʔɔ:.ní:t]]		
742		<i>aoníít</i>		
743		[ao-nii]-t		
744		[hole-by.needle.TI]-2SG.IMP		
745		‘pierce it!’ [FR 301]		
746				
747	b. After C		c. After V	
748		[[ʔâ:.kɔ(:).ni.mə]]		[[ʔiç.ta.wɔ:.nja:.kjoʔ.pa]]
749		<i>áakaonima</i>		<i>iihtawáóniaakio’pa</i>
750		aak-[ao-nii]-wa		ii\oht-a-[ao-niaki]-o’p-a
751		FUT-[hole-by.needle.TI]-3		IC\INS-IPFV-[hole-by.needle.AI]-21.IND-3
752		‘she will pierce it’ [FR 301]		‘can opener, bradawl’ [FR 32]
753				
754	Abstracting away from the predictable [ʔ] that occurs at the left edge of the verbal			
755	complex, the second set of roots begins in a long non-high vowel at the left edge, and			
756	with an initial [w] after a vowel, (23a–c). In other words, these roots exhibit a Ø ~ [w]			
757	alternation at the left edge of the root. There are no roots in Blackfoot which begin in a			
758	[w] at the left edge and after a vowel, (23d).			
759				
760	(23)	Left edge	After C	After V
761	a.	[[ʔa:n-]]	[[an-]]	[[wa:n-]]
762	b.	[[ʔɛ: ^s t-]]	[[ɛ(:) ^s t-]]	[[wɛ: ^s t-]]
763	c.	[[ʔɔ:-]]	[[ɔ(:)-]]	[[wɔ:-]]
764	d.	*[[w...]]	*[[Ø...]]	*[[w...]]
765				
				Gloss
				‘tell’
				‘towards’
				‘hole’

Logically, the $\emptyset \sim [w]$ alternation is compatible with two analyses: either the roots begin in an underlying long vowel and the $[w]$ is epenthesized in some contexts, or the roots begin in an underlying $/w/$ which is deleted at the left edge and after a consonant. The traditional analysis in Frantz (2017:177, Rule 15) is that these roots begin with an underlying $/w/$ which is deleted, and this analysis is adopted in later research (Elfner 2006:91; Bliss 2013:152). However, there are several arguments which show that the epenthesis analysis must be correct, which I return to below.

5.2.3 $[j]$ glide after a prefix

For the third group of roots, the root begins with long $[i:]$, $[o:]$, or $[a:]$ at the left edge and after a vowel, and $[j]$ following $[a]$, (24)–(27). For example, *yiistom-* ‘body’ begins with a long $[i:]$ at the left edge, (24a), a long $[i:]$ after a consonant, (24b), and $[ji:]$ after the vowel $[a]$, (24c). The $/k/$ at the end of *aak-* ‘FUT’ regularly assibilates to $[ks]$ before $[i]$, (24b), unless the following consonant is $[t]$, as in (17b) and (25b) (Chávez Peón 2015).

(24) a. Left edge

$[[\text{ʔi}:\text{.}^{\text{s}}\text{tóm.ʔs}:\text{.kox}^{\text{w}}.\text{sít}]]$
iistómsskoohsit
 $[\text{yiistom} - \text{ssko}] - \text{ohsi} - \text{t}$
 $[\text{body} - \text{by.body.TA}] - \text{REFL.AI} - 2\text{SG.IMP}$
 ‘exercise!’ [FR 314]

b. After C

$[[\text{ʔá}:\text{.ksi}:\text{.}^{\text{s}}\text{tóm.ʔs}:\text{.kox}^{\text{w}}.\text{si.má}]]$
áaksiistómsskoohsima
 $\text{aak} - [\text{yiistom} - \text{ssko}] - \text{ohsi} - \text{m} - \text{a}$
 $\text{FUT} - [\text{body} - \text{by.body.TA}] - \text{REFL.AI} - \text{IND} - 3$
 ‘she will exercise’ [FR 314]

c. After V

$[[\text{ʔá.ji}:\text{.}^{\text{s}}\text{tom.ʔs}:\text{.kox}^{\text{w}}.\text{si.wá}]]$
áyiistomsskoohsiwa
 $\text{a} - [\text{yiistom} - \text{ssko}] - \text{ohsi} - \text{wa}$
 $\text{IPFV} - [\text{body} - \text{by.body.TA}] - \text{REFL.AI} - 3$
 ‘she is exercising’ [FR 314]

(25) a. Left edge

$[[\text{ʔi}:\text{.tsi.max.ka:t}]]$
iitsimaahkaat
 $[\text{yiit} - \text{imaa}] - \text{hkaa} - \text{t}$
 $[\text{storage} - \text{AI}] - \text{GET.AI} - 2\text{SG.IMP}$
 ‘store food for the winter!’ [FR 315]

b. After C

c. After V

802	[[ʔâ:.ki:.tsi.max.ka:.wə]]	[[ʔi:.tá.ji:.tsi.max.kəʔ.pi]]
803	<i>áakiitsimaahkaawa</i>	<i>íitáyiitsimaahkao'pi</i>
804	aak-[yiit-ímaa]-hkaa-wa	ii'it-a-[yiit-ímaa]-hkaa-o'p-i
805	FUT-[storage-AI]-GET.AI-3	IC\LOC-IPFV-[storage-AI]-GET.AI-21.IND-IN.SG
806	'store food for the winter!' [FR 315]	'August (lit. when we prepare food
807		for storage)' [FR 38]

808
809 (26) a. Left edge

810 [[ʔô:.mit]]
811 *óomit*
812 [yoom-i]-t
813 [husband-AI]-2SG.IMP
814 'marry!' [FR 320]
815

816 b. After C

817 [[ʔâ:.kô:.mi.wə]]
818 *áakóomiwa*
819 aak-[yoom-i]-wa
820 FUT-[husband-AI]-3
821 'she will marry' [FR 320]
822

c. After V

817 [[ʔá.ka:.jô:.mi.wə]]
818 *ákaayóomiwa*
819 akaa-[yoom-i]-wa
820 PRF-[husband-AI]-3
821 'she's married' [FR 320]
822

823 (27) a. Left edge

824 [[ʔa:.mo.jiʔ.po.jit]]
825 *aamoyíi'poyit*
826 [yaam-oyi-i'poyi]-t
827 [twisted-mouth-speak.AI]-2SG.IMP
828 'joke/jest from a twisted mouth!' [FR 310]
829

830 b. After C

831 [[ʔâ:.ka:.mo.jiʔ.po.ji.wə]]
832 *áakaamoyíi'poyiwa*
833 aak-[yaam-oyi-i'poyi]-wa
834 FUT-[twisted-mouth-speak.AI]-3
835 'he will joke' [FR 310]
836

c. After V

831 [[ʔá.ja:.mo.jiʔ.po.ji.wə]]
832 *áyaamoyíi'poyiwa*
833 a-[yaam-oyi-i'poyi]-wa
834 IPFV-[twisted-mouth-speak.AI]-3
835 'he is joking' [FR 310]
836

837 Abstracting away from the predictable [ʔ] that occurs at the left edge of the verbal
838 complex, the third set of roots begins in a long vowel at the left edge and after a
839 consonant, but with an initial [j] after a vowel, (28a–d). In other words, these roots
840 exhibit a Ø ~ [j] alternation at the left edge of the root. No roots that begin in a long mid
841 vowel [ɛ:] or [ɔ:] at the left edge of the verbal complex exhibit a Ø ~ [j] alternation,

(28d,e).¹³ This gap probably has a diachronic explanation (see Section 8.2). There are no roots in Blackfoot which begin in a [j] at the left edge and after a vowel, (28f).¹⁴

(28)	Left edge	After C	After V	Gloss
a.	[[ʔi: ^s tom-]]	[[i: ^s tom-]]	[[ji: ^s tom-]]	‘body’
	[[ʔi:t-]]	[[i:t-]]	[[ji:t-]]	‘food storage’
b.	[[ʔo:m-]]	[[o:m-]]	[[jo:m-]]	‘husband’
c.	[[ʔa:m-]]	[[a:m-]]	[[ja:m-]]	‘twisted’
d.	*[[ʔε:. . .]]	*[[ε:. . .]]	*[[jε:. . .]]	
e.	*[[ʔɔ:. . .]]	*[[ɔ:. . .]]	*[[jɔ:. . .]]	
f.	*[[j. . .]]	*[[∅. . .]]	*[[j. . .]]	

Just as with the second group of roots, the $\emptyset \sim [j]$ alternation in (28) are logically compatible with either an epenthesis or deletion analysis. Unlike the $\emptyset \sim [w]$ alternation, there are several factors which support the traditional analysis, where the roots begin in an underlying /j/ which is deleted at the left edge and after a consonant. In the next section I turn to an analysis of underlying forms for all three groups of roots.

5.3 Arguments for underlying forms

There are three groups of roots above. Group 1 roots begin in a long vowel at the left edge and after a prefix. Group 2 roots exhibit a $\emptyset \sim [w]$ alternation at the left edge of the root. Group 3 roots exhibit a $\emptyset \sim [j]$ alternation at the left edge of the root. In this section I present several arguments that Group 2 roots begin in an underlying long vowel and that [w] is epenthesized in certain contexts, but Group 3 begin in an underlying /j/ which is deleted in certain contexts.

¹³ An anonymous reviewer points out that the gaps in (28d,e) support the analysis of long /ε:/ and /ɔ:/ as true monophthongs rather than diphthongs /ai/ and /ao/. Because there are roots which begin in [ja:...] after a vowel, if the mid vowels were /a/-initial diphthongs then we would also expect roots like [jai...] and [jao...] to exist. They do not, which suggests that the mid vowels must have some other type of representation which does not begin in /a/.

¹⁴ Some roots have a nasal [m] or [n] at the left edge but a [j] after a vowel (Weber 2020:143ff), and there may be other partially overlapping patterns as well, not shown here.

Groups 1 and 2 are in complementary distribution when we consider the quality of the first vowel of the root. Together, the two sets exhaust the inventory of long vowels which I argued for in Section 4. This is shown in (29) with Group 1 in (29a) and Group 2 in (29b). This points toward an analysis where both sets of roots have the same type of underlying representation. Group 1 in (29a) begins in a long vowel in all positions, so the most parsimonious analysis is that they begin in an underlying long vowel. Therefore, Group 2 in (29b) most likely does as well. Those underlying representations are given in the fourth column in (29). Notice that with this reanalysis some roots begin in underlying long mid vowels, which is to be expected if they are phonemic.

(29) Groups 1 and 2: Roots with initial long vowels

	Left edge	After C	After V	UR	Gloss
a.	[[ʔi:ts:k-]]	[[i:ts:k-]]	[[i:ts:k-]]	/i:ts:k-/	‘scuffle’
	[[ʔo:k-]]	[[o:k-]]	[[o:k-]]	/o:k-/	‘bead’
b.	[[ʔa:n-]]	[[an-]]	[[wa:n-]]	/a:n-/	‘tell’
	[[ʔe:ˢt-]]	[[ε(:)ˢt-]]	[[we:ˢt-]]	/ε:ˢt-/	‘towards’
	[[ʔɔ:-]]	[[ɔ(:)-]]	[[wɔ:-]]	/ɔ:-/	‘hole’
c.	*[[w...]]	*[[Ø...]]	*[[w...]]		

In contrast, Group 3 overlaps in distribution with Groups 1 and 2 when we consider the quality of the first vowel of the root. Some roots in Group 3 begin in [i:] and [o:], (30a,b), just like Group 1, (29a), while other roots in Group 3 begin in [a:], (30c), just like Group 2, (29b). The roots in Group 3 themselves do not begin in a natural class of long vowels: they include high and low vowels, but not the mid vowels [ε:] and [ɔ:]. This points toward an analysis where the roots in Group 3 have a different kind of underlying representation than in Groups 1 and 2. If Groups 1 and 2 begin in a long vowel, then Group 3 must begin with an underlying /j/. Those underlying representations are given in the fourth column in (30).

(30) Group 3: Roots with initial /j/ before long vowels

899		Left edge	After C	After V	UR	Gloss
900	a.	[[ʔi: ^s tom-]]	[[i: ^s tom-]]	[[ji: ^s tom-]]	/ji: ^s tom-/	‘body’
901		[[ʔi:t-]]	[[i:t-]]	[[ji:t-]]	/ji:t-/	‘food storage’
902	b.	[[ʔo:m-]]	[[o:m-]]	[[jo:m-]]	/jo:m-/	‘husband’
903	c.	[[ʔa:m-]]	[[a:m-]]	[[ja:m-]]	/ja:m-/	‘twisted’
904	d.	*[[ʔɛ:. . .]]	*[[ɛ:. . .]]	*[[jɛ:. . .]]		
905	e.	*[[ʔɔ:. . .]]	*[[ɔ:. . .]]	*[[jɔ:. . .]]		
906	f.	*[[j. . .]]	*[[∅. . .]]	*[[j. . .]]		
907						

908 Alternatives to the above analyses do not explain the facts as neatly. If Group 3 roots
909 in (30) begin with underlying long vowels and epenthesize a [j] after prefixes, then there
910 is no explanation for the gaps in (30d,e). If roots begin in /ɛ:/ and /ɔ:/, then why do those
911 not also epenthesize [j] after vowels? The roots in Groups 1 and 2 in (29a,b) would also
912 be expected to epenthesize [j], unless they had some other underlying representation,
913 such as beginning with an underlying /w/.

914 If we take the analysis of Group 3 above to be correct, but assume that Groups 1 and
915 2 begin with an underlying /w/, then essentially we are saying there are no roots that
916 begin in an underlying long vowel – a strange lexical gap! In order to account for Group
917 1, the /w/ would have to delete specifically between /a/ and a long high vowel in order to
918 feed vowel coalescence. There are no surface instances of *[wi], and other alternations
919 prove that we would expect a sequence [ji:] from underlying /wi:/. However, sequences
920 like [...aji:...] and [...awo:...] are allowed in Blackfoot. This means there is no
921 phonotactic reason for a [w] to delete in these environments.

922 Finally, if we take the analysis of Groups 1 and 3 to be correct, but assume that Group
923 2 begins in /w/ (essentially the analysis in Frantz 2017), this would mean that roots begin
924 in long high vowels, but no roots begin in long non-high vowels – another strange lexical
925 gap. There is another argument against taking Group 2 roots to begin in /w/. The reason is
926 that [w] is restricted even further by the vowel to the left. The [w] only occurs before
927 long non-high vowels after [a] but not after [i] or [o].¹⁵

¹⁵ There is a question about whether [w] appears between [o] and a root that begins in a long non-high vowel. There are not many, if any, such examples in Frantz and Russell (2017). Although Weber

For example, the stem *mokaki* ‘wise’ occurs before the long [a:] at the left edge of *aan-* ‘tell’ in (31). There is no [w] at the left edge of the root in (31), and instead the final [i] of *mokaki* ‘wise’ becomes a glide. This process of glide formation occurs before all of the other vowels other than [i] as well. Therefore, an analysis that treats /w/ as part of the underlying form of *aan-* ‘tell’ must resort to ad-hoc deletion of /w/ after /i/ in (31). Example (32a) shows that *mokaki* ‘wise’ is an independent stem that ends in [i], while (32b) repeats (20b) to show that *sap-* ‘correct’ has a short vowel after a consonant.

(31) [[ʔâ:.ko.ka.kja:.ni.^stsi:.wá.ji]]
áakokakiaanistsiiwáyí
 aak–mokaki–[aan–ist]–ii–w=ayi
 FUT–wise–[tell–TA]–3SUB–3=OBV.SG
 ‘he will advise him’ [FR182]

(32)	a.	[[ʔâ:.ko.ka.ki.wə]]	b.	[[sa.pa.ní. ^s tsís]]
		<i>áakokakiwa</i>		<i>sapanístsisa</i>
		aak–[mokaki]–wa		sap–[aan–ist]–:s
		FUT–[wise.AI]–3		correct–[tell–TA]–2SG:3.IMP
		‘she will be smart’ [FR182]		‘respond correctly to him!’ [FR 241]

The root *kissto-* ‘without apparent cause’ occurs before the long [a:] at the left edge of *aanat-* ‘pretty’ in (33). There is no [w] at the left edge of the root in (33), and instead the final [o] of *kissto-* ‘without apparent cause’ deletes. Therefore, an analysis that treats /w/ as part of the underlying form of *aan-* ‘tell’ must resort to ad-hoc deletion of /w/ after /o/ in (33). This process of deletion occurs before the other two non-high long vowels as well. Example (34a) shows that *kissto-* truly ends in [o], while (34b) shows that *aanat-* ‘pretty’ has a short vowel allomorph after a consonant just like *aan-* ‘tell’ in (33b).

(33) [[ʔé:.ksis.tâ:.nət.tsi:.ji]]
áiksisstáanattsiiyi
 a–kissto–[aanat–tsii]–yi
 IPFV–without.cause–[pretty–by.heat.II]–IN.SG

(2020:136) gives a putative example, the glide is actually the final segment of the prefix rather than epenthetic. I will assume that epenthesis does not occur in this context.

960 ‘electric light, flashlight’ [FR8]

961

- 962 (34) a. [[ʔé:.ksis.tó:.ma.to.max.ka:.wə]] b. [[ʔâ:.ka.nət.tsi:.wə]]
 963 *aiksisstoomatomaahkaawa* *áakanattsiwa*
 964 a-ksissto-[omat-omaahkaa]-wa aak-[aanat-tsii]-wa
 965 IPFV-no.cause-[start-drive.AI]-3 FUT-[pretty-by.heat.II]-3
 966 ‘automobile’ [FR8] ‘it will light up’ [FR289]
 967

968 As the examples above demonstrate, the contexts where [w] occurs at the left edge of
 969 a root are quite narrow. In particular, [w] only occurs before long non-high vowels, and
 970 only when the preceding vowel is [a], summarized in Table 3. An analysis where [w] is
 971 epenthesized in these few contexts is far more parsimonious than an analysis where the
 972 /w/ is underlying but deletes in each of the other vowel contexts, after a consonant, and at
 973 the left edge of the verbal complex. Under this analysis, [w]-epenthesis emerges as one of
 974 several competing strategies to resolve vowel hiatus, along with coalescence, glide
 975 formation, and deletion. (See Elfner 2006 for an overview of vowel hiatus resolution
 976 strategies in Blackfoot.)

977

978 **Table 3:** Vowel contexts where [w] occurs at the left edge of roots

V1\V2	i:	o:	a:	ɛ:	ɔ:
i	×	×	×	×	×
o	×	×	×	×	×
a	×	×	✓	✓	✓

979

980 To summarize, roots that begin in long vowels at the left edge have one of two types
 981 of underlying forms. Some roots begin in an underlying long vowel (Groups 1 and 2),
 982 with epenthetic [w] in the onset of a heavy syllable between two non-high vowels. Other
 983 roots begin in an underlying /j/ (Group 3), which deletes at the left edge of the word and
 984 after a consonant. Other possible analyses are less parsimonious or create strange gaps in
 985 the root lexicon. One remaining question is why there are no roots that begin in /j/
 986 followed by a long mid vowel. A possible answer is that /ɛ:/ and /ɔ:/ are recent
 987 innovations in Blackfoot, which I discuss more in Section 8.2.

5.4 Summary

In summary, this section argued that there are no lexical gaps, and that some roots begin in underlying long mid vowels, as expected if those vowels are phonemic. I presented patterns of alternation for roots that begin in long vowels at the left edge of the verbal complex. Roots fall into three groups based on the form of the allomorph after a prefix: in this position, a root may begin in a long vowel, begin in [w], or begin in [j]. Ultimately I argued that these patterns result from two kinds of underlying representation: roots begin with an underlying long vowel or with a palatal glide /j/, but never with a /w/. Although [w] occurs at the left edge of some roots in non-initial contexts, it is best analyzed as an epenthetic glide which occurs between a short non-high vowel and a long non-high vowel. The traditional analysis takes these roots as beginning in an underlying /w/, meaning there would be no roots in the lexicon which begin with a long non-high vowel, which would be an odd gap if those vowels are phonemic. Under my analysis, roots begin in all five long vowel qualities, providing additional support for phonemic /ε:/ and /ɔ:/. In the next section I raise supporting evidence for this analysis via a small corpus analysis of the dictionary (Frantz and Russell 2017).

6 Dictionary corpus results

In Section 4 I argued that the long mid vowels /ε:/ and /ɔ:/ are contrastive in Blackfoot in morpheme-internal positions. In Section 5 I considered patterns of alternation for roots that begin with vowels or glides and argued that roots can begin in all five long vowels or the palatal glide /j/, but that no roots begin in an underlying labiovelar glide /w/. This is somewhat surprising, given that previous accounts explicitly analyze some roots as beginning in a labiovelar glide /w/ which deletes in initial position (Elfner 2006:91; Bliss 2013:152; Frantz 2017:177). Additionally, with such targeted datasets it is difficult to know whether all patterns have been represented without a larger corpus study (see discussion in Andersson 2024). Perhaps there are other roots which begin in glides in both positions (initial and non-initial) or which begin in [w] before high vowels which were accidentally missed.

In this section I examine a small corpus consisting of the set of entry headers in Frantz and Russell (2017), the most recent print dictionary of Blackfoot. This dictionary lists entries under both <w> and <y>, implicitly adopting the analysis that some verbs begin in underlying /w/ and /j/. I show that the roots listed under vowels, <w>, and <y> are not evenly distributed between nouns and verbs, and that the entries under the glides <w> and <y> are predominantly verb stems. I then argue that the way that Frantz and Russell (2017) determine verb stem entry headers ensures that the verb stems listed under <w> and <y> are the non-initial stem variant – the same position where epenthetic [w] occurs. In other words, the dictionary corpus supports my analysis above, where [w] only occurs as an epenthetic glide between vowels.

6.1 Methods

The Frantz and Russell (2017) dictionary was digitized by the Blackfoot Lab at Yale as part of the Blackfoot Words lexical database project (Weber et al. 2023). The first author of the dictionary shared the original Word document (.docx) format, where each entry was formatted with particular delimiters and annotations. An R script was used to pull information from this document into a text document with tab separated values.¹⁶ This was then uploaded to Google Sheets and double-checked for accuracy by research assistants as part of the procedures for Blackfoot Words described in Weber et al. (2023). I extracted a subset of the data in this document: (1) the dictionary entry header, (2) the translation, and (3) the morphological type and subclassification (e.g., the part of speech).

The corpus contains 5,180 entries in total, including some entries which appear in the Word document but not in the published dictionary. I grouped the entry headers by morphological type and subclassification into one of four simplified categories: “noun”, “verb”, “adjunct”, and “other”. The “noun” and “verb” categories are noun stems and verb stems, respectively. The “adjunct” category follows the terminology from Frantz and Russell (2017) and Frantz (2017). These are morphological preverbs and are not to be confused with syntactic adjuncts. This class is heterogenous and consists of roots and

¹⁶ Thank you to Sarah Babinski for writing this script.

grammatical prefixes which combine with other roots, stems, or finals to create new stems. Crucially, many of these can combine with either nominal or verbal stems, though there is a large set of adjuncts that combine with verbs only. “Other” is a catch-all category for the remaining entries. Table 4 gives the correspondences between the original abbreviations from Frantz and Russell (2017:*xvi–xvii*) and my simplified categories.

Table 4: Correspondences between abbreviations in Frantz and Russell (2017) and the simplified categories

Morphological type and subclassification	Abbreviation	Simplified category
noun, animate	nan	noun
noun, inanimate	nin	noun
noun, animate relational	nar	noun
noun, inanimate relational	nir	noun
noun, unknown animacy	n_n	noun
noun, either animacy	nan/nin	noun
verb, animate intransitive	vai	verb
verb, inanimate intransitive	vii	verb
verb, transitive animate	vta	verb
verb, transitive inanimate	vti	verb
verb, inanimate intransitive	vi	verb
verb, intransitive, unknown animacy	v_i	verb
verb, intransitive, either animacy	vai/vii	verb
verb roots	vrt	verb
adjuncts	adt	adjunct
uninflected	und	other
medial	med	other
final	fin	other
demonstrative	dem	other
pronoun	pro	other
suffix	suf	other

The entry headers are alphabetized into the following 13 sections: a, h, i, k, m, n, o, p, s, t, w, y, '. There are fewer sections than there are phonemic contrasts in Blackfoot, so some of the sections contain entry headers that begin with multiple segments from the inventory. For example, the <a> section lists entry headers that begin in short <a> [a], long <aa> [a:], long <ai> [ɛ:], and long <ao> [ɔ:].

In the next section I consider the distribution of entry headers across the 13 sections.

6.2 Results

I show in this section that although there are dictionary entries listed under <w>, they are all non-initial variants of verbs which begin in a heavy syllable containing a non-high vowel. In other words, the entry headers happen to be the glide-initial allomorphs of roots that begin in a non-high vowel. This lends support to my argument that no verbs begin in an underlying /w/. The entries listed under <y> are also non-initial variants of verbs but begin with long [a:], long [i:], or long [o:]. This means that they overlap in distribution with verbs that begin in underlying long vowels, lending support to my argument that verbs may begin in an underlying /j/.

6.2.1 Entries under vowels and glides by category

Table 5 shows the distribution of entry headers across all sections divided by category. Since the focus is on vowels and glides, I have collapsed all of the other consonants into a single line. The second column lists all of the orthographic (di)graphs in each section that correspond to distinct phonemes. Except for doubled graphs, which correspond to long sounds, I have included phonetic symbols in square brackets wherever those differ from the orthographic symbols.

Table 5: Number of entry headers listed under vowels, glides, and consonants by simplified category

Section	Entries begin with	noun	verb	adjunct	other	Total
<a>	a, aa, ai [ɛ:], ao [ɔ:]	354	82	53	26	515
<i>	i, ii	413	957	187	39	1596
<o>	o, oo, oi	222	591	80	21	914
<w>	w	1	274	37	2	314
<y>	y [j]	4	178	16	3	201
C	h, k, m, n, p, s, ss, t, ' [ʔ]	804	590	174	72	1640
Grand Total		1798	2672	547	163	5180

Notably, there are very few nouns listed under <w> (only one) and <y> (only four).¹⁷ All five nouns have a related entry listed under a vowel or <m>. For example, *iihkawa* ‘neighbor/associate/companion’ and *yiihkowa* ‘associate’ are clearly the same stem listed in two different sections. The noun *yiitsittsimaan* ‘meat (cut) for drying’ is also listed under <i> as *iitsittsimaan* ‘cut/slice meat (thinly) for drying’. Two of the entries even include verbiage that suggests the two entries are variants of one another. The entry *aakókaa’tsis* ‘Sundance camp’ is listed as a “variant of” *waakokaa’tsis*, and the entry *iaapan* ‘blood’ is listed as a “variant of” *aaapan* ‘blood’. The entries *miistapipaitapiiyi* and *yiiistapipaitapiiyi* ‘ancestor’ are both analyzed as containing the morpheme *miistap-* ‘away’. I take this evidence to mean that no nouns begin in a glide, and that the five listed in Table 5 are variants of other entries.

Why are verbs but no nouns listed under <w> and <y>? The answer has to do with how the entry headers were determined. As Frantz and Russell (2017:xx–xxi) explains, the entry headers for nouns and stative verbs come from unprefix stems (with inflection removed), but the entry headers for non-stative verbs come from the form of the verb in a non-initial position (typically after the durative prefix *á-*). Consequently, non-stative verbs are the only type of entry listed under their non-initial variants. The reason why only verbs are listed under <w> is because roots only begin with <w> in non-initial positions, and only verb entries are created from non-initial variants. This is indirect support for my analysis that [w] is epenthetic in non-initial positions.

There is a small but notable number of adjuncts listed under <w> and <y> as well. As mentioned above, adjuncts combine primarily with verb stems or verbalizing derivational suffixes. It stands to reason that adjuncts attached to non-stative verb stems would be analyzed in the same fashion as non-stative verb stems.

The following sections examine the entries listed under <w> and <a> more closely. I find that the first vowel of the stem is always non-high in these sections, and is either long or in a closed syllable. As discussed in Section 5.3, [w] is epenthesized exactly

¹⁷ I treated entries that began with <iV(V)> as <yV(V)>. Some entries begin with VVV, which I treated as a long vowel <VV>.

before this set of environments, which I take as further evidence that the <w> at the left edge of the root is not underlying.

6.2.2 Entries under <w> by syllable shape

Within the <w> section there are further distributional gaps. As shown in Table 6, all <w> entries are followed by a select few sequences: a long non-high vowel in an open syllable <aa> [a:], <ai> [ɛ:], or <ao> [ɔ:]; a low vowel followed by a cluster <a'C> or <aahC>; or a low vowel followed by a geminate <aCC>.¹⁸ Once syllabified, these are all sequences which are parsed to a heavy CVV or CVC syllable. There are also 12 entries with a short vowel <a> followed by a single C, which are parsed into a light CV syllable. Eleven of these are followed by [j], and Weber (2020:128ff) argues that the sequence [aj] patterns with heavy syllables as if it were a diphthong. Several generalizations can be stated here. First, the vowel immediately following <w> is never high. Second, the <w> is always the onset to a heavy syllable which contains a long vowel, is closed, or contains an [aj] diphthong. There is only a single exception to this generalization.

Table 6: Number of entries listed under <w> by following context and simplified category

Section: <w>	noun	verb	adjunct	other	Total
waa...	1	182	21		204
wai...		9	2	1	12
wao...		9			9
wa'C...		27	3	1	31
wahC...					0
waahC...		25	7		32
waaCC...		4	1		5
waCC...		9			9
waC...		9	3		12
Grand Total	1	274	37	2	314

¹⁸ In this table, the <wai> line includes entries that are followed by Ø, by a single <C>, or by a <'C> cluster. The <wao> line includes entries that are followed by Ø, or a single <C>.

The obvious comparison set are entries under the <a> vowel section. As shown in Table 7, <a> entries begin with long <aa> [a:], <ai> [ɛ:], or <ao> [ɔ:].¹⁹ There are also entries with a short <a> in a closed syllable, followed by a cluster <a'C> or <ahC> or <aahC>,²⁰ or a geminate <aCC>. Unlike the <w> section of the dictionary, there are many entries which begin in a short vowel in a light syllable <aC>, although most of these are nouns. Some of the verbs listed in the <a> section are bound at the left edge (“adt. req.”) and most of the remainder are listed as the “initial variant of” an entry that begins in <w>.

Table 7: Number of entries listed under <a> by following context and simplified category

Section: <a>	noun	verb	adt	other	Total
aa...	87	18	9	3	117
ai...	66	7	4	1	78
ao...	38	1	1	1	41
a'C...	21	35	3	2	61
ahC...		4	1		5
aahC...	5		6		11
aCC...	2	1	4	14	21
aC...	135	16	25	5	181
Grand Total	354	82	53	26	515

What seems to be happening here is that verbs which begin with a non-high vowel are listed under <w> only if that vowel is long or will otherwise be parsed to a heavy syllable, but are listed under <a> if the vowel is short and will be parsed to a light syllable. Nouns are uniformly listed under <a>. There are some exceptions either way,

¹⁹ In this table, the <ai> line includes entries that are followed by ∅, a <'C> cluster, or a <CC> geminate. The <ao> line includes entries that are followed by ∅, a <'C> cluster, or a single <C>, a <CC> geminate, or an <hC> cluster. The <a'C> line includes entries that are followed by ∅, a <'>, or a <'C> cluster.

²⁰ In Frantz's (1978, 2017) orthography, a single vowel before <hC> denotes a devoiced vowel and a doubled letter before <hC> denotes a short voiced vowel. Although Frantz (2017) frequently using the spelling <ahC> in word-initial position, the word-initial syllable is never pronounced with a devoiced vowel before <hC>.

but these appear to be inconsistencies in how noun and verb entry headers are created or else are due to entries which are simply “(initial) variants of” another entry.

6.3 Summary

The corpus study confirms the generalization from Section 5 that [w] at the left edge of roots is highly restricted and is best analyzed as an epenthetic element. First, the entries listed under <w> and <y> are nearly entirely verbs, due to the way that entry headers for all non-stative verbs are created from non-initial variants of the root or stem (typically following the durative prefix *á-*). Nouns and stative verbs undergo the same types of alternations, but because their entry headers are created from initial allomorphs, these are listed under <a>. Second, root-initial <w> is only ever followed by a non-high vowel which is parsed into a heavy syllable (e.g., either the vowel is long, or it is followed by a cluster or geminate). Taken together, this means that the <w> at the beginning of an entry occurs before a non-high heavy syllable and immediately after <a> – the exact same conditions where [w] is epenthesized. Given that this is the same generalization found in Section 5.3, the corpus results help prove that the root alternation data was not cherry-picked but is truly representative of the lexicon.

Crucially, the corpus results support the findings in Section 5 that some roots begin in underlying long mid vowels, as expected if those vowels are phonemic. The evidence from distribution, root alternations, and the corpus study supports the inclusion of /ɛ:/ and /ɔ:/ in the inventory, increasing the number of long vowels in Blackfoot from three to five. This raises the question of how two vowel phonemes went undiscovered in Blackfoot until now, which I turn to next in Section 7.

7 Undiscovered phonemes in Blackfoot

The long mid vowels /ɛ:/ and /ɔ:/ are clearly contrastive phonemes in Blackfoot, and yet they have thus far been entirely missed. In this section I explore some of the factors that may have led to this. I also compare the previous arguments made about long mid vowels to the types of arguments made in this paper. Broadly speaking, previous arguments were made on the basis of economy and predictability, without considering the full range of

distributional facts examined in this paper. The distributional generalizations only became possible to make once enough facts about allophony and morphology in Blackfoot were understood.

One thread of research in Frantz (1978, 2017) on Blackfoot phonology concentrates on grouping allophones into families of phonetically similar sounds, without considering how these sounds interface with morphology or whether they contrast with other phonemic categories. With respect to the long mid vowels, Frantz (1978:312) explicitly points out that there are two families of non-contrastive sounds consisting of positional and free variants. One family consists of [eː] ~ [ɛ] ~ [æː] ~ [ai], while the second consists of [ɔː] ~ [ɔ] ~ [ao]. His main concern is in choosing a phonemic representation for both families. He recognized that the short monophthongs only occurred before long consonants and treated the remaining phones as free variants, though in later work he recognized that [ai] and [ao] mainly occur before glottal stops (Frantz 2017:2–3; described in Section 4.3.1). He chooses the diphthongs /ai/ and /ao/ as the best phonemic representation of these sounds due to economy: the diphthongs are composed of two vowel qualities which occur independently (Frantz 1978:312; 2017:186), whereas the long monophthongs are not, and can be easily treated as allophonic variants. An alternative argument (pursued in Section 4.3.1) would be to take the allophones with the least restricted distribution as the phonemic category. There was no clear “elsewhere” allophone at the time Frantz (1978) was written, which may have had some influence on Frantz’s argumentation, but there was by the time the first edition of the Frantz (2017) grammar was written. Still, it is clear that the evidence from economy is at odds with the evidence from distribution.

Other researchers focus more strongly on these vowels’ relationship to morphology rather than their phonetic realization. Linguists have long understood that the long mid vowels arise predictably across morpheme boundaries. A few researchers failed to notice the tautomorphemic long mid vowels entirely, and clearly state that diphthongs and “contracted” (coalesced) vowels *only* occur across morpheme boundaries (Taylor (1969:32, 86–88; Kinsella 1972:20). When every known instance of these allophones is

predictable, then of course there is no reason to posit separate phonemes. It makes sense how this could happen: the unpredictable cases of /ɛ:/ and /ɔ:/ are very low in frequency and there are few minimal pairs in Blackfoot in general (Derrick 2007, note 2, estimates about 300 in total). In addition, because much remains unknown about the morphemic analysis inside Blackfoot stems, researchers could very easily conclude that all instances of [ɛ:] and [ɔ:] indicate the presence of a boundary.

The only research I know of which explicitly discusses tautomorphemic cases of long mid vowels is in Weber (2020:42–43), Author and Coauthor (2024:273–276), and Author and Coauthor (2025). Prior to these works, other researchers focus on long mid vowels at boundaries without either denying the tautomorphemic cases nor pointing out that they exist (Elfner 2006:95ff; Frantz 2017:186). Frantz interprets both long mid vowels as underlying sequences of vowels because that is clearly their origin across morpheme boundaries. Given his extensive knowledge of the language, I think it is safe to conclude that he meant this analysis should extend to tautomorphemic cases. This analysis maximizes economy, because new phonemes do not need to be added to the inventory, and both of the vowels in the underlying sequence are independently needed. It also integrates an assumption from generative phonology that the same rules which account for morpheme alternations also frequently account for allophony. In this particular case, the underlying sequence never surfaces (except perhaps in the form of a diphthong before a glottal stop), but the implication is that a learner of the language would be able to extend the analysis from alternations to the tautomorphemic cases.

This paper relied primarily on arguments of distribution rather than economy. Crucially, in Section 4 I established that tautomorphemic long mid vowels contrast with other mid vowels via overlapping distribution in immediate local contexts. This required a morphological analysis of each word to ensure that /ɛ:/ and /ɔ:/ were contained within morphemes. There are also differences in distribution between /ɛ:/ and /ɔ:/ and the diphthong [oi], which truly only arises in morphologically derived environments. I took this as a strong indication that /ɛ:/ and /ɔ:/ have a different type of representation than [oi], and I argued for underlying long monophthongs based on the distribution of

positional allophones. In Section 5 I considered patterns of root alternation. The fact that [ɛ:] and [ɔ:] have the same pattern of [w]-epenthesis as long [a:] indicates that these three vowels pattern as a class. The fact that the patterns of alternation of the non-high long vowels complement the patterns of alternation of the high long vowels supports this view as well. Finally, in Section 6 I pointed out distributional gaps in the existing analysis by evaluating entry headers in the current dictionary. Here, it was striking that nearly no nouns are listed under the <w> section of the dictionary, and also that <w> can only be followed by a few vowel qualities and syllable shapes. These gaps are more easily explained under an analysis where the entries under <w> begin with phonemic long vowels. These distributional arguments do not necessarily *refute* a generative analysis that derives long mid vowels from underlying vowel sequences. Such an analysis would generate all the correct surface strings. However, the distributional facts that I have outlined here would remain coincidental. Considering that learners do indeed rely on distributional knowledge, I argue that my analysis does a better job at integrating the distributional facts into the phonological grammar.

Finally, there are a few other structural factors in this particular case that may have also contributed to why the long mid vowels remained “undiscovered” until recently.

1. These vowels are in an “intermediate” phonological relationship (to use the terminology of Hall 2013), because they are not entirely predictable (allophonic) in all environments, nor are they entirely unpredictable (contrastive). This is the reason that multiple analyses are possible: an underlying vowel sequence (based on economy) or underlying long monophthongs (based on distribution). Note that technically the other long vowels also have an intermediate phonological relationship, since long vowels also arise when two identical short vowels occur across a morpheme boundary. However, the long mid vowels seem less robustly contrastive for the following reasons as well.
2. If phonemes are determined in part by their structural position in the inventory, then /ɛ:/ and /ɔ:/ are less robustly phonemic than the other long vowels because they have no short counterparts /ɛ/ and /ɔ/. This is a case of “partial” contrast,

where only a subset of a natural class exhibits a contrast (e.g., Austin 1988). This also has the effect of making minimal pairs harder to discover. Minimal pairs had been established for other short and long vowels, e.g. (1a), and it is likely that if the mid vowels also had distinctive length that they would have been established via minimal pairs with short mid vowels. Instead, contrast must be established by establishing contrast with other long vowels.

3. There is “a high degree of variability” (Hall 2013:239ff) in how /ɛ:/ and /ɔ:/ are produced for any particular speaker, dialect, and phonological context. In contrast, the other long vowels have fairly stable phonetic targets and do not vary between monophthong and diphthong pronunciations. This contributes to the long mid vowels feeling more marginal and less robustly contrastive.

One overarching theme throughout the history of Blackfoot research is how difficult it can be to determine contrasts in understudied and polysynthetic languages. Because there are few minimal pairs, the most straightforward argument for contrast (lexical distinctiveness) is difficult to use. As discussed above, before the major positional allophones had been determined it was impossible to make an argument for the underlying form of each family of allophones. Even today, there are many open questions about speaker variation in how these sounds are pronounced (as discussed in Section 4.3.1). Similarly, before the morphological analysis of many words has been determined it was impossible to see that some long mid vowels are tautomorphemic. This is not at all trivial – the morphological analysis inside of stems in particular is still not well understood. In Section 8.3 I describe a lexical database which is being developed for the analysis of Blackfoot derivation, which can be used in the future to study other morphophonological patterns in Blackfoot.

In summary, there are multiple factors that contributed to hiding two phonemic contrasts in Blackfoot. There are very few minimal pairs, the unpredictable instances are fairly infrequent, and until recently our knowledge of Blackfoot allophony and morphology was not detailed enough to make the kinds of distributional arguments I made in this paper. Additionally, the types of argumentation in previous work were not

based on distribution but on reasons of economy. One takeaway from this case study is that researchers of polysynthetic languages may need to privilege distributional tests in order to determine contrast. Overlapping distribution can only be assumed if two sounds occur in the same syllabic and morphological context.

8 Summary and discussion

In this paper I reanalyzed the Blackfoot vowel and glide inventory based on arguments of phonological distribution in static contexts and root alternations as well as distributional gaps in a corpus of dictionary entry headers. I argued that the vowel inventory includes two additional long vowels, /ɛ:/ and /ɔ:/, and that the labiovelar glide [w] that occurs at the left edge of some roots is epenthetic. The result is an asymmetric vowel inventory with five long vowels against three short vowels.

The case study of Blackfoot shows that some phonemic contrasts can go undiscovered for a long time in polysynthetic languages, in part because there are often few minimal pairs and some aspects of the phonological and morphological grammar may not be fully known. An additional problem for the Blackfoot long mid vowels is that the vowel contrasts are in an “intermediate phonological relationship”, and that the unpredictable instances of these vowels are infrequent compared to the predictable instances. Although there are several possible analyses of these facts, my argument in this paper is that the distributional evidence comes down in favor of treating the unpredictable instances as contrastive phonemes rather than analyzing them in the same way as predictable cases. This highlights the importance of distributional evidence when conducting phonemic analyses of polysynthetic languages and my hope is that the phonological and corpus methods shown here can be adapted to other languages without many minimal pairs.

There are several points of discussion. In Section 8.1 I consider some of the synchronic implications for Blackfoot empirically, but also in terms of linguistic theory and typology. Section 8.2 discusses some aspects of Blackfoot historical phonology. Finally, Section 8.3 explores implications for current and forthcoming lexical resources.

8.1 Synchronic implications

The reanalysis in this paper adds two phonemes, /ɛ:/ and /ɔ:/, to the Blackfoot inventory, and recasts root-initial [w] as an epenthetic glide. This reanalysis leads to interesting consequences for phonological typology across inventories (Section 8.1.1). In the future, several targeted acoustic studies could corroborate this research (Section 8.1.2).

8.1.1 Typology of vowel inventories

A consequence of this analysis is that Blackfoot has an asymmetrical vowel inventory, with a greater number of long vowels than short vowels. This type of vowel inventory is potentially expected under dispersion theory (Flemming 2004), because long vowels are more perceptible, and can carry more contrasts in F1 and F2. There are also other languages where only a subset of vowels exhibit a length contrast, suggesting that asymmetrical inventories may be more common than previously thought, and even expected as one way for languages to encode more contrasts.

Within Algonquian, Nishnaabemwin (Valentine 2001:34–41), Oji-Cree (Slavin 2012), and Plains Cree (Wolfart 1973:79) each have three short vowel qualities /i a o/ but four long vowels /i: e: a: o:/, where the long /e:/ has no phonemic short counterpart. (But see Russell 1992, who suggests that /e:/ in Plains Cree is actually a heavy diphthong with the underlying representation /ai/, fused by a late, possibly phonetic rule which is separately attested in external sandhi process in the language.) Goddard (1979:96) reconstructs Proto-Eastern-Algonquian to have four long vowels, *a:, *e:, *i:, *o:, and two short vowels, *a and *e; in most modern Eastern Algonquian languages the length contrast has been reanalyzed as a contrast between full and reduced vowels.

Outside of Algonquian, vowel length in Papapana is contrastive only for the three front vowels, /i/, /e/, /a/, and not for the back vowels /ɔ/ and /u/ (Smith-Dennis 2021). Lithuanian has an asymmetrical vowel system where only the high and low vowels contrast for length, and mid vowels are only long (Klimas 1970; Mathiassen 1996; Campos-Astorkiza 2012). Future typological studies could reveal how common such systems are and what their properties are.

8.1.2 Acoustic studies

Although this study focused on phonological patterns, the data raise several questions about the phonetic properties of Blackfoot. At a basic level, future studies should explore the variation in vowel quality for the long mid vowels across different phonological contexts. A recent articulatory description of Káínai Blackfoot (based on a single speaker) includes words with long mid vowels, but the dataset is not systematic and lacks several important phonological environments – for example there are only three words with a long mid vowel before a glottal stop (Author and Coauthor 2025). Future studies might focus on acoustic analyses in corpora (such as the preliminary study in Author and Coauthor 2024), or create targeted wordlists to study acoustic realizations systematically across phonological environments and speakers.

There are also several interesting morphological contexts that should be considered. For example, it is not known whether the tautomorphological cases of long mid vowels have the same range of phonetic variation as the predictable cases. Since tautomorphological long /ɛ:/ and /ɔ:/ are relatively rare, these types of studies will likely also require targeted elicitation. As discussed in Section 5.2.2, one important environment would be roots that begin in a long mid vowel after consonants. The prediction based on root alternations is that long mid vowels pattern with variable length [a:] and shorten in this environment. Another interesting set of environments would be vowel hiatus contexts. In the reanalysis presented in this paper, [w] is only epenthesized between two non-high vowels when the second one occurs in a heavy syllable. This is just one of many possible hiatus resolution strategies, including coalescence (of two short vowels), glide formation (of a high front vowel [i] before a long vowel), and more. Elfner (2006) and Weber (2020) present different analyses based primarily on orthography, but this is again an area that would benefit from acoustic studies to understand how those orthographic representations manifest.

8.2 Diachronic implications

The long mid vowels in Blackfoot must be innovated, because Proto-Algonquian (PA) only had four long vowels (*i̯, *e̯, *a̯, and *o̯), and did not contain diphthongs

(Bloomfield 1946; Goddard 1979). At least some cases of long mid vowels may be reflexes of other long monophthongs. The root (underlined) in PA **re·hr·e·wa* ‘he breathes’ has a reflex containing [ɛ:] in one Blackfoot dialectal variant: *%saiit-ami-t* ‘breathe!’ (listed alongside another dialectal variant *%siit-* in Frantz and Russell 2017:238). In this word PA **e·* > B *%ai* [ɛ:] in one dialect, although the usual reflex of PA **e·* is B *ii* (Berman 2006:266). If this change is regular in other forms as well, then Blackfoot [ɛ:] may have arisen from a “primary split” in the phoneme **e·* (Hoenigswald 1965), where PA **e·* either became B [ɛ:] or merged with the existing reflex of PA **i·*. However, this idea would require further substantiation and also would not explain how [ɔ:] was innovated. Another possible source would be sequences like PA **ahi* and **aho*. Intervocalic PA **h* deleted in Blackfoot (Goddard 2018:84), which would create B **ai* and **ao* diphthongs. Considering that older sources often write <ai> and <ao> or something like a diphthong (see Weber et al. 2023:1221–1222), it is possible that Blackfoot once had two phonemic diphthongs /ai/ and /ao/ which have since fused to long mid vowels, similar to the argument in Russell (1992) for Plains Cree discussed in the previous section. Further studies of Blackfoot historical phonology might shed light on how the long mid vowels were innovated.

Another question is how Blackfoot developed /j/-initial roots, as no roots with initial **y* existed in Proto-Algonquian (e.g., Bloomfield 1946; Hewson 1993). The root alternations from (28) are repeated below in (42), where it is striking that /j/ only occurs before reflexes of the four Proto-Algonquian long vowels. Although PA **i·* and **e·* merged to Blackfoot [i:], Blackfoot **t* only pre-assibilates to [ʰt] after **i·* and not after **e·*. The first vowel in /ji:stom-/ ‘body’ must be from PA **i·* because the following **t* is pre-assibilated to [ʰt], and the first vowel in /ji:t-/ ‘food storage’ must be from PA **e·* because the following **t* is not pre-assibilated. Since PA **o·* > B [o:] and PA **a·* > B [a:], all four PA long vowels occur in /j/-initial roots in Blackfoot.

1391

1392	(42) Roots with initial /j/ before long vowels and PA vowel reconstructions					
1393	Left edge	After C	After V	UR	PA	Gloss
1394	a. [ʔi:stom-]	[i:stom-]	[ji:stom-]	/ji:stom-/	< *i·	‘body’

1395		[[ʔi:t-]]	[[i:t-]]	[[ji:t-]]	/ji:t-/	< *e·	‘food storage’
1396	b.	[[ʔo:m-]]	[[o:m-]]	[[jo:m-]]	/jo:m-/	< *o·	‘husband’
1397	c.	[[ʔa:m-]]	[[a:m-]]	[[ja:m-]]	/ja:m-/	< *a·	‘twisted’
1398							

1399 One hypothesis is that root-initial /j/ comes from an archaic form of initial change
1400 which prefixed *ay- to long vowels in the initial syllable (Costa 1996). Blackfoot
1401 contains remnants of such a system in roots that begin in a consonant (Taylor 1967). If
1402 the *y in the prefix were reanalyzed as part of the following root, we would expect root-
1403 initial [j] to occur only before the reflexes of the PA long vowels, as in (42). This also
1404 means that the long mid vowels must have been innovated relatively recently, after initial
1405 change had been lexicalized as part of the root.

1406 The analysis in this paper also refutes Berman’s (2006:270) claim that verb stems in
1407 PA *a· > B *waa*. Instead, root-initial [w] is only epenthesized under narrow conditions.
1408 The process of [w]-insertion might be a reflex of the process of <y> insertion between
1409 two long vowels in Proto-Algonquian (Bloomfield 1946:93), though with a narrowed
1410 scope because [w]-insertion only occurs between a subset of vowels.

1411 8.3 Lexical resource development

1412 Finally, I touch on some of the implications this research has for lexical resources. Within
1413 the dictionary (Frantz and Russell 2017; Genée and Russell 2018), it would make sense
1414 to create new sections in the dictionary for <ai> and <ao> which are separate from <a>,
1415 to emphasize that these represent different phonemes.²¹ The entries that are currently
1416 under <w> could also be merged with the remaining entries under <a>, though this would
1417 require the dictionary to use entry headers that are *not* created from non-initial stems.
1418 One option would be to use the imperative for non-statives and to use the third person
1419 singular forms for statives. The sections play a much larger role in the print version of the
1420 dictionary (Frantz and Russell 2017) rather than the online version (Genée and Russell
1421 2018), but both versions of the dictionary utilize entry headers.

²¹ These vowels are indeed taught as separate sounds to learners of the language, as in the sound chart on the Mioohpokoiksi learning hub (<https://sites.google.com/view/mioohpokoiksi/appendices/sounds-of-blackfoot>).

The Blackfoot Words database is a resource created in part to more easily find patterns in root alternations like the ones in this paper (Weber et al. 2023). This is a relational database of inflected words, stems, and morphemes in Blackfoot which have been digitized from over 30 sources, representing all four major dialects, and spanning the years 1743–2017. Version 1.1 of the database includes lexical forms from nine grammars, dictionaries, and wordlists. Frantz and Russell (2017) is not included in Version 1.1 but will be included in a future release. The idea is that Frantz & Russell (2017) contains many gaps, where roots are not always shown in all three relevant positions (the left edge, after a consonant, and after a vowel). By including multiple sources, Blackfoot Words is able to fill many of those gaps. The inflected words in the database are also being analyzed into constituent stems and roots, which provides information about the morphological context of each root. These factors mean that in the future it will be easier to search for root alternations in Blackfoot and to make further arguments about the lexicon and phonological inventory.

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